Probability and Stochastic Processes

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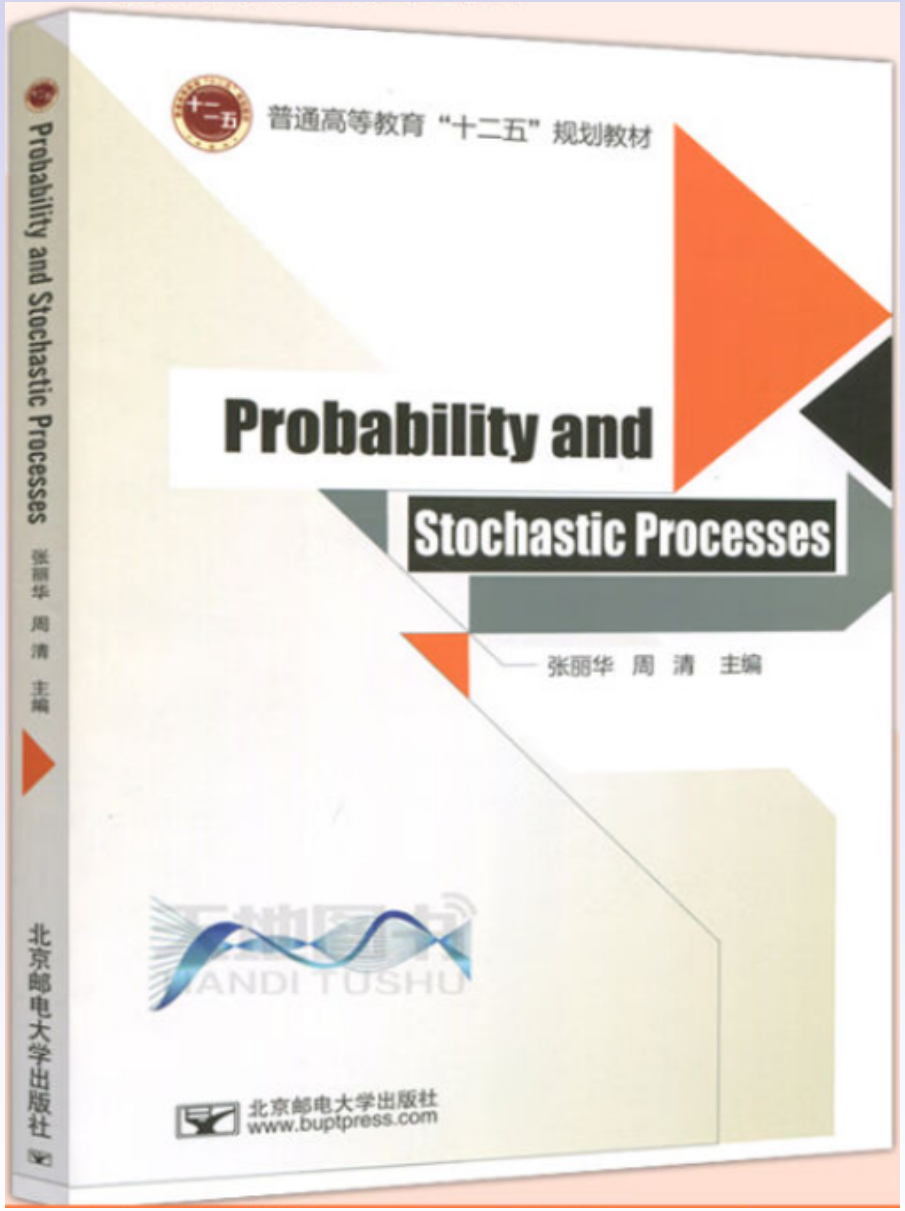
Evaluation

- A comprehensive final exam (70%)
- Attendance (10%)
- Homework (10%)–Submit homework every Monday
- Midterm (10%)–Write summary

Textbook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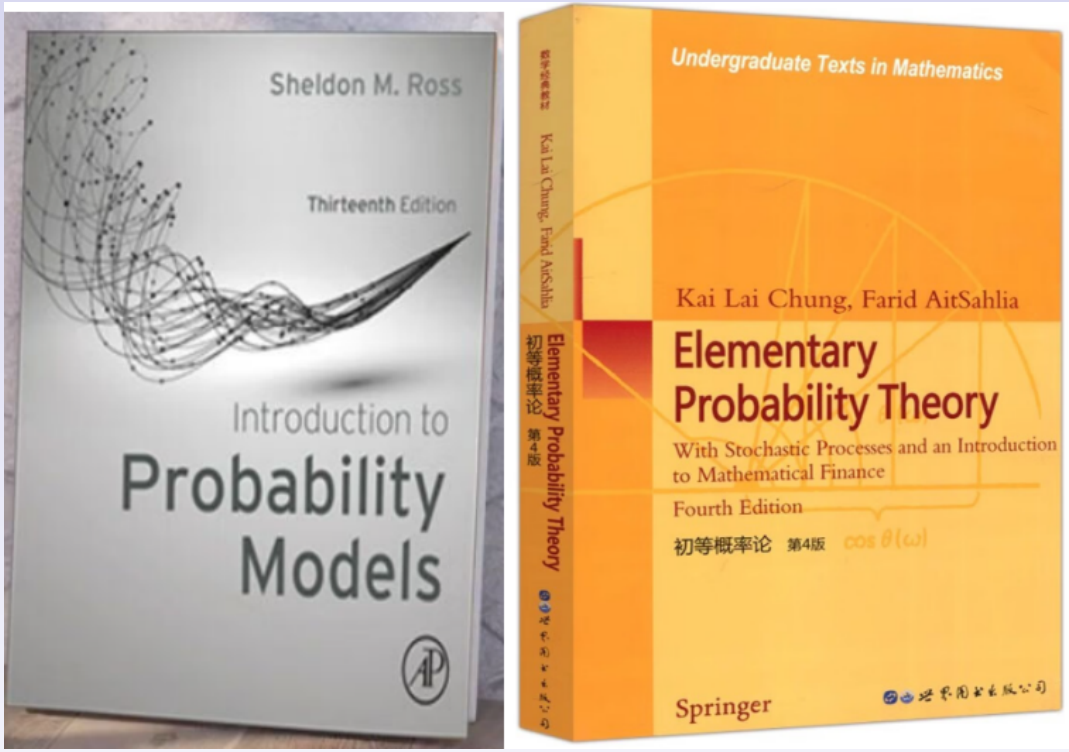
References

Figure: https://www.doc88.com/p-84661319000453.html



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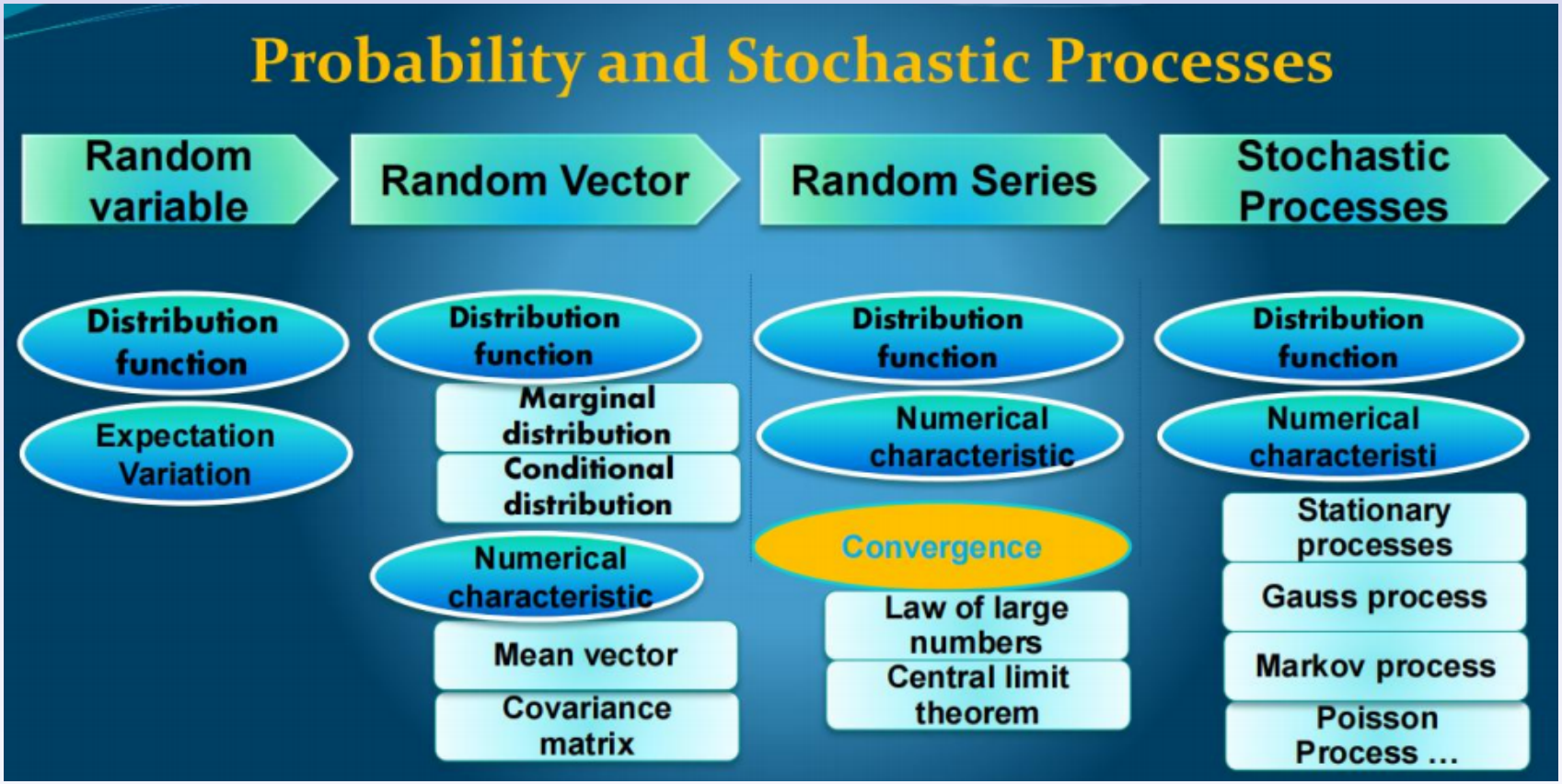
Probability and Stochastic Processes
Random variable
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Distribution function
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Conditional distribution
Numerical characteristic
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Chapter 3. Random Variables

Chapter 4. Sequences of Random Variables

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Chapter 1. Events and Their Probabilities

- Section 1.1 The History of Probability
- Section 1.2 Experiment, Sample Space and Random Event
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Section 1.2 Experiment, Sample Space and Random Event

- 1.2.1 Basic Definitions
- 1.2.2 Events as Sets

1.2.1 Basic Definitions

A random phenomenon is a situation in which we know what outcomes could happen, but we do not know which particular outcome did or will happen.

'Random' in statistics is not a synonym for 'haphazard' but a description of a kind of order that emerges only in the long run.

1.2.1 Basic Definitions

Random experiment usually has the following three characteristics.

(i) Repeatability: it can be repeated under the same conditions.

(ii) Predictability: it has more than one outcome and we know all possible outcomes before the experiment.

(iii)Uncertainty: the outcome of the experiment will not be known in advance.

In this book, we shall abbreviate "random experiment" to experiment and denote it by E.

1.2.1 Basic Definitions

Some examples of random experiment:

E_1 : Determination of the sex of a newborn child.

E_2 : Roll a die and observe which number appears.

E_3 : Flip two coins and observe the outcomes.

E_4 : Roll two dice and observe the outcomes.

E_5 : Observe call times for a call center.

E_6 : Measure the lifetime of cars.

How to record the experiment data?

Each possible outcome: ***sample point*** (ω)
The set of all possible outcomes: ***sample space*** (Ω)

1.2.1 Basic Definitions

Example

E_1 : *Determination of the sex of a newborn child.*

For E_1, $\Omega_1 = \{g, b\}$, where the outcome g means that the child is a girl and b that it is a boy.

Example

E_2 : *Roll a die and observe which number appears.*

For E_2, $\Omega_2 = \{1, 2, 3, 4, 5, 6\}$, where the outcome i means that i appeared on the die, $i = 1, 2, 3, 4, 5, 6$.

Example

E_3 : *Flip two coins and observe the outcomes.*

For E_3, $\Omega_3 = \{(H, H), (H, T), (T, H), (T, T)\}$, where H means head and T means tail.

1.2.1 Basic Definitions

Example

E_4 : *Roll two dice and observe the outcomes.*

For E_4,

$$\Omega_4 = \begin{pmatrix} (1,1) & (1,2) & (1,3) & (1,4) & (1,5) & (1,6) \\ (2,1) & (2,2) & (2,3) & (2,4) & (2,5) & (2,6) \\ (3,1) & (3,2) & (3,3) & (3,4) & (3,5) & (3,6) \\ (4,1) & (4,2) & (4,3) & (4,4) & (4,5) & (4,6) \\ (5,1) & (5,2) & (5,3) & (5,4) & (5,5) & (5,6) \\ (6,1) & (6,2) & (6,3) & (6,4) & (6,5) & (6,6) \end{pmatrix}$$

where the outcome (i, j) is said to occur if i appears on the first die and j appears on the second die, $i, j = 1, 2, \cdots, 6$.

1.2.1 Basic Definitions

Example

E_6 : *Measure the lifetime of cars.*

For E_6, $\Omega_6 = [0, \infty)$, where the outcome t is the lifetime of a car, $0 \leqslant t < \infty$.

- Any subset of the sample space Ω is known as an ***random event*** or ***event***. Events are usually denoted by capital letters $A, B, C, \cdots$.
- Those events must occur in the experiment are called the ***inevitable events***.

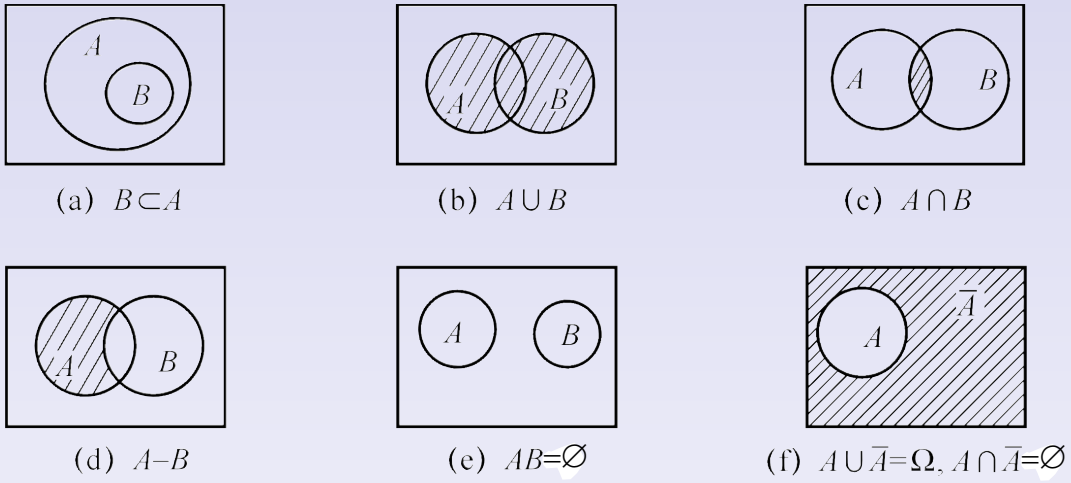
  Sample space Ω is an inevitable event.
- Those could not happen anytime are said to be ***impossible events***. We usually denote impossible events by $\emptyset$.

Section 1.2 Experiment, Sample Space and Random Event

- 1.2.1 Basic Definitions
- 1.2.2 Events as Sets

1.2.2 Events as Sets

Figure: The relationships between events A and B



$$\bigcup_{i=1}^{n} A_i \triangleq A_1 \cup A_2 \cup \cdots \cup A_n, \quad \bigcup_{i=1}^{\infty} A_i \triangleq A_1 \cup A_2 \cup \cdots$$

$$\bigcap_{i=1}^{n} A_i \triangleq A_1 \cap A_2 \cap \cdots \cap A_n, \quad \bigcap_{i=1}^{\infty} A_i \triangleq A_1 \cap A_2 \cap \cdots$$

1.2.2 Events as Sets

Table: The jargons in set theory and probability theory

notation	Set jargon	Probability jargon
Ω	Collection of objects	Sample space
ω	Member of Ω	Elementary event, outcome
A	Subset of Ω	Events that some outcome in A occurs
A or $\overline{A}$	Complement of A	Event that no outcome in A occurs
$A \cap B$	Intersection	Both A and B
$A \cup B$	Union	Either A or B or both
$A - B$	Difference	A, but no B
$A \subseteq B$	Inclusion	If A, then B
$\emptyset$	Empty set	Impossible event
Ω	Whole space	Certain set

1.2.2 Events as Sets

Let A, B and C be the random events of experiment E. The operations of the events will satisfy the following rules:

(i) Commutatively $\quad A \cup B = B \cup A,\ AB = BA.$

(ii) Associatively $\quad A \cup (B \cup C) = (A \cup B) \cup C = A \cup B \cup C.$

$$A(BC) = (AB)C = ABC.$$

(iii) Distributively $\quad A \cup (B \cap C) = (A \cup B) \cap (A \cup C),$

$$A(B \cup C) = (AB) \cup (AC).$$

(iv) De Morgan's law $\quad \overline{A \cup B} = \overline{A} \cap \overline{B},\ \overline{A \cap B} = \overline{A} \cup \overline{B}.$

More generally, $\overline{(\bigcup_i A_i)} = \bigcap \overline{A_i},\ \overline{(\bigcap_i A_i)} = \bigcup \overline{A_i}.$

1.2.2 Events as Sets

In addition, there are some common properties, such as

(1) $\overline{\overline{A}} = A$.

(2) $A \cup B \supset A$ and $A \cup B \supset B$. In particular, if $A \subset B$, then $A \cup B = B$.

(3) $A \cap B \subset A$ and $A \cap B \subset B$. In particular, if $A \subset B$, then $A \cap B = A$.

(4) $A - B = A - AB = A\overline{B}$.

(5) $A \cup B = A \cup \overline{A}B$.

1.2.2 Events as Sets

Example

Suppose that A, B and C are three events and $D = \{$at least one of the three events will occur$\}$. Try to describe the event D by events A, B and C.

Solution. We describe the event D by three different ways:

(i) directed method: $D = A \cup B \cup C$,

(ii) decomposition method:

$D = A\overline{B}\ \overline{C} \cup \overline{A}B\overline{C} \cup \overline{A}\ \overline{B}C \cup \overline{A}BC \cup A\overline{B}C \cup AB\overline{C} \cup ABC$,

(iii) inverse method:

$D = \overline{\overline{D}} = \overline{\{A,\ B \text{ and } C \text{ will not occur}\}} = \overline{\overline{A}\ \overline{B}\ \overline{C}}$.

1.2.2 Events as Sets

Example

Select an integer from 1 to 1000 at random. Calculate the probability of the event that the integer is not divisible by 6 and 8.

Section 1.3 Probabilities Defined on Events

- 1.3.1 Classical Probability
- 1.3.2 Geometric Probability
- 1.3.3 The Frequency Interpretation of Probability

1.3.1 Classical Probability

- Definition of Classical Experiment
- Definition of Classical Probability
- Examples

Classical Probability

Definition of Classical Experiment

The classical definition or interpretation of probability is identified with the works of Jacob Bernoulli and Pierre-Simon Laplace. It is based on the concept of equally likely outcomes.

Classical Probability

Definition of Classical Experiment

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Generally, a random experiment E is ***classical*** if

1. E contains only different **limited** basic events, that is,

$$\Omega = \{\omega_1, \omega_2, \cdots, \omega_n\}.$$

   We call this kind of sample space ***simple space***, and

2. all outcomes are **equally** likely to occur.

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   We call this kind of sample space ***simple space***, and

2. all outcomes are **equally** likely to occur.

For example, when a fair coin is tossed, there are two possible outcomes: a head or a tail. A head and a tail are equally likely to occur.

Classical Probability

Definition of Classical Probability

For classical random experiment E, the corresponding problems of probability belong to the category of ***classical probability***.

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Definition

For classical random experiment E, $\Omega = \{\omega_1, \omega_2, \cdots, \omega_n\}$, we define the probability of event A as

$$P(A) = \frac{\#A}{\#\Omega} \tag{1}$$

where $\#A$ means the number of all possible outcomes of event A, $\#\Omega$ means the number of all possible outcomes of sample space Ω. If A is comprised by k different elementary events, then

$$P(A) = \frac{k}{n}.$$

Classical Probability

From above definition of classical probability, the probability of impossible event is defined as

$$P(\emptyset) = 0. \tag{2}$$

Classical Probability

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$$P(\emptyset) = 0. \tag{2}$$

Theorem

For classical random experiment E, *the probability has the following properties:*
(i) *for every event* A, $P(A) \geqslant 0$,
(ii) $P(\Omega) = 1$,
(iii) *for every finite sequence of* n *disjoint events* $A_1, A_2, \cdots, A_n$,

$$P(\bigcup_{i=1}^{n} A_i) = \sum_{i=1}^{n} P(A_i).$$

Property (iii) *is called finite additivity.*

Classical Probability

Proof. (i) and (ii) are obvious.
Now let us prove (iii). Suppose that

$$A_i = \{s_1^i, s_2^i, \cdots, s_{k_i}^i\},$$

where $i = 1, 2, \cdots, m$ and $s_l^i \neq s_r^i, 1 \leqslant l, r \leqslant k_i, l \neq r$.
We know that $P(A_i) = \dfrac{k_i}{n}$ from equation (1.1). It is obvious that

$$\bigcup_{i=1}^{m} A_i = \{s_1^1, s_2^1, \cdots, s_{k_1}^1; s_1^2, s_2^2, \cdots, s_{k_2}^2; \cdots; s_1^m, s_2^m, \cdots, s_{k_m}^m\}.$$

Since $A_1, A_2, \cdots$ and A_m are disjoint events, these elementary events must be different. Then we have

$$P(\bigcup_{i=1}^{m} A_i) = (\sum_{i=1}^{m} k_i)\Big/n = \sum_{i=1}^{m} \frac{k_i}{n} = \sum_{i=1}^{m} P(A_i).$$

$\square$

Classical Probability

Example

Consider the dice-roll experiment E_4 and calculate the probability of each of the possible values of the sum of the two numbers that may appear.

Recall: E_4 : Roll two dice and observe the outcomes.

Classical Probability

Example

Consider the dice-roll experiment E_4 and calculate the probability of each of the possible values of the sum of the two numbers that may appear.

Recall: E_4 : Roll two dice and observe the outcomes.

Solution. From Example 1.2.4, we know $\#\Omega = 36$. It can be easily found that the probability of each of the outcomes in E_4 is $1/36$. Let A_i denote the event of the sum of the two numbers is i, $i = 2, 3, \cdots, 12$. For A_2, the only outcome in Ω for which the sum is 2 is the outcome $(1, 1)$. Therefore,

$$P(A_2) = \frac{\#A_2}{\#\Omega} = 1/36.$$

Classical Probability

By continuing in this manner, we obtain the following probability for each of the possible values of the sum:

$$P(A_2) = P(A_{12}) = \frac{1}{36}, \quad P(A_3) = P(A_{11}) = \frac{2}{36},$$

$$P(A_4) = P(A_{10}) = \frac{3}{36}, \quad P(A_5) = P(A_9) = \frac{4}{36},$$

$$P(A_6) = P(A_8) = \frac{5}{36}, \qquad P(A_7) = \frac{6}{36}.$$

$\square$

Classical Probability

Example

Suppose that a urn contains α white balls and β black balls, and that $m + n$ balls are to be chosen at random at one time $(m \leqslant \alpha, n \leqslant \beta)$. We shall determine the probability that exactly m white balls will be chosen.

Classical Probability

Example

Suppose that a urn contains α white balls and β black balls, and that $m+n$ balls are to be chosen at random at one time ($m \leqslant \alpha, n \leqslant \beta$). We shall determine the probability that exactly m white balls will be chosen.

Solution. Since $m+n$ balls are to be chosen at random at one time,

$$\sharp\Omega = \binom{\alpha+\beta}{m+n},$$

and each of these $\binom{\alpha+\beta}{m+n}$ possible combinations is equally probable.

Let A be the event that contains exactly m white balls and n black balls. Now let us find $\sharp A$.

Classical Probability

When a combination of m white balls and n black balls is formed, the number of different combinations in which m white balls can be selected from the α available balls is $\binom{\alpha}{m}$, and the number of different combinations in which β black balls can be selected from the β available balls is $\binom{\beta}{n}$. Since each of these combinations of m white balls can be paired with each of the combinations of n black balls to form a distinct sample, the number of combinations containing exactly m white balls is $\binom{\alpha}{m}\binom{\beta}{n}$.
Therefore, the desired probability is

$$P(A) = \frac{\#A}{\#\Omega} = \frac{\binom{\alpha}{m}\binom{\beta}{n}}{\binom{\alpha+\beta}{m+n}}.$$

$\square$

Example

Suppose that an urn contains α white balls and β black balls, and that $k+1$ $(k+1 \leqslant \alpha+\beta)$ balls are to be selected consecutively at random without replacement. Try to determine the probability that the last selected ball is exactly white ball.

Solution. Let A be the event that the last selected ball is exactly white ball.

$$P(A) = \frac{\#A}{\#\Omega} = \frac{\alpha\binom{\alpha+\beta-1}{k}k!}{\binom{\alpha+\beta}{k+1}(k+1)!} = \frac{\alpha}{\alpha+\beta}.$$

It is worth noting that the probability has nothing to do with the value of k. $\square$

Classical Probability

Example

Suppose that there are n people, each person will be assigned to any of the $N(n \leqslant N)$ rooms with the same probability $1/N$. We shall determine the probabilities of the following events:

A : For the given n rooms, there is exactly one person in one room.

B : There is exactly one person in one room.

C : There are exactly m people in a given room.

Classical Probability

Example

Suppose that there are n people, each person will be assigned to any of the $N(n \leqslant N)$ rooms with the same probability $1/N$. We shall determine the probabilities of the following events:
A : For the given n rooms, there is exactly one person in one room.
B : There is exactly one person in one room.
C : There are exactly m people in a given room.

Solution. In this experiment, the number of all different possible outcomes are N^n.
For event A, since n rooms has been given in advance, the first people has n choices, the second people has $n-1$ choices, $\cdots$.
So $\sharp A = n!$. Then

$$P(A) = \frac{n!}{N^n}.$$

Classical Probability

For event B, the number of the outcomes that n rooms are to be chosen at random from N rooms is $\binom{N}{n}$. The next step is the same as in event A, then

$$P(B) = \frac{\binom{N}{n} n!}{N^n}.$$

For event C, the first step is choosing m people from n people at random and let them in the given room, and the number of this process is $\binom{n}{m}$. The second step is that $n - m$ people are about to be assigned $N - 1$ rooms in any way and the number of this process is $(N-1)^{n-m}$. So the probability of event C is

$$P(C) = \frac{\binom{n}{m}(N-1)^{n-m}}{N^n}.$$

$\square$

Classical Probability

Example

(**Birthday problem**) If a group consists of n people, what is the probability that at least two of them have the same birthday? Ignore leap years and assume that each day in the year is equally likely as a birthday.

Classical Probability

Example

(**Birthday problem**) If a group consists of n people, what is the probability that at least two of them have the same birthday? Ignore leap years and assume that each day in the year is equally likely as a birthday.

Solution. Let us consider the sample space of all ordered lists of n birthdays. One such list assigns a birthday to each of the n people. All simple events in this sample space are equally likely to occur. Consider the event

$$B = \{\text{at least two people have the same birthday}\}.$$

Classical Probability

To compute $P(B)$, we first compute $P(\overline{B})$. The complement of B is the event

$$\overline{B} = \{\text{no two people have the same birthday}\}.$$

Its probability is given by

$$P(\overline{B}) = \frac{365 \cdot 364 \cdots (365 - n + 1)}{365^n}.$$

Therefore,

$$P(B) = 1 - P(\overline{B}) = 1 - \frac{365 \cdot 364 \cdots (365 - n + 1)}{365^n}.$$

It can be easily found that the probability reaches 100% when the number of people reaches 367 (since there are 366 possible birthdays, including February 29). However, 99% probability is reached with just 57 people, and 50% probability with 23 people. This means that if there are 23 or more people in a room, there is more than 1 chance out of 2 that at least two people have the same birthday! $\square$

Classical Probability

It is important that we do not confuse the above birthday problem with the following one. If a group of n people is in a room, what is the probability that at least one person in the group has the same birthday as you. In this case, let

$$B = \{\text{at least one person has the same birthday as you}\}.$$

We obtain in this case

$$P(B) = 1 - \left(\frac{364}{365}\right)^n.$$

The smallest value of n where $P(B) > 1/2$ is now $n = 253$ which is much larger than 23.

Classical Probability

Example

A reception has received 12 visits in a week. Suppose that all 12 receptions are proceeded on Tuesday and Thursday. Is the reception time required?

Classical Probability

Example

A reception has received 12 visits in a week. Suppose that all 12 receptions are proceeded on Tuesday and Thursday. Is the reception time required?

Solution. Assume that the reception time is not specified. Then, the probability of the event that all 12 receptions are proceeded on Tuesday and Thursday is

$$\frac{2^{12}}{7^{12}} = 0.0000003.$$

This is a very small probability. Practical experience shows that rare event should seldom occur (referred to as the impossible principle). Now in this example, the rare events have happened in one experiment. So there is a reason to doubt the validity of the assumption and we can reach the conclusion that the reception time is required. □

Section 1.3 Probabilities Defined on Events

- 1.3.1 Classical Probability
- 1.3.2 Geometric Probability
- 1.3.3 The Frequency Interpretation of Probability

Geometric Probability

Example

what is the probability that a point is selected randomly from the interval $[0, 1]$? If your answer is 0, how can you get that? Is it coming from $\frac{1}{\infty}$? If so, what is the probability of choosing the interval $[0, 0.5)$? Is your answer $\frac{1}{2}$? Obviously, we can not get that by classical probability.

What is the mean area of the polygonal regions formed when randomly oriented lines are spread over the plane?

What is the chance that three random points in the plane form an acute (rather than obtuse) triangle?

Geometric Probability

Example

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What is the mean area of the polygonal regions formed when randomly oriented lines are spread over the plane?

What is the chance that three random points in the plane form an acute (rather than obtuse) triangle?

The definition of classical probability can not be used any more when the concept of equally likely outcomes is about to be extended to a line or 2D or 3D area.

Geometric Probability

A random experiment E is called to be ***geometric*** if

(i) the sample space is a measurable (such as length, area, volume, etc.) region, i.e., $0 < L(\Omega) < \infty$, and
(ii) the probability of every event $A \subset \Omega$ is proportional to the measure $L(A)$ and has nothing to do with its position and shape.

In this case, we define the probability of event A as

$$P(A) = \frac{L(A)}{L(\Omega)}$$

and $P(\emptyset) = 0$.

Geometric Probability

Theorem

For Geometrical random experiment E, *the probability has the following properties:*
(i) *for every event* A, $P(A) \geqslant 0$,
(ii) $P(\Omega) = 1$, *and*
(iii) *for every countable disjoint events* $A_1, A_2, \cdots$,

$$P\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} P(A_i).$$

Property (iii) *is called countable additivity.*

Geometric Probability

Proof. (i) and (ii) are obvious.
(iii) According to countable additivity in measure theory, i.e., for every countable disjoint sequence $A_1, A_2, \cdots$,

$$L\Big(\bigcup_{i=1}^{\infty} A_i\Big) = \sum_{i=1}^{\infty} L(A_i),$$

then

$$P\Big(\bigcup_{i=1}^{\infty} A_i\Big) = \frac{L\big(\bigcup\limits_{i=1}^{\infty} A_i\big)}{L(\Omega)} = \sum_{i=1}^{\infty} \frac{L(A_i)}{L(\Omega)} = \sum_{i=1}^{\infty} P(A_i).$$

$\square$

Geometric Probability

Example

(**Lunch date problem**) You and one of your friends arrange to meet between 12:00 and 13:00. As a result, it is possible for one of you to arrive at random between 12:00 and 13:00 and waits exactly 20 minutes for another one. After 20 minutes, one of you leaves if another person has not arrived. What is the probability that you and your friend will meet?

Geometric Probability

Example

(**Lunch date problem**) You and one of your friends arrange to meet between 12:00 and 13:00. As a result, it is possible for one of you to arrive at random between 12:00 and 13:00 and waits exactly 20 minutes for another one. After 20 minutes, one of you leaves if another person has not arrived. What is the probability that you and your friend will meet?

Solution. Let x and y be the time you and your friend arriving the gate, respectively, then our sample space is a square $\Omega = \{(x, y)|\ 12 \leqslant x, y \leqslant 13\}$. Let M be the event that two of you will meet at the gate, then M will occur if and only if $|x - y| \leqslant 1/3$, i.e.,

$$M = \{(x, y)|\ 12 \leqslant x, y \leqslant 13,\ |x - y| \leqslant 1/3\}.$$

Geometric Probability

In this problem, the expression "equally likely to occur" means that the probability that the sample point is located in a special region $M \subset \Omega$ is proportional to the area of M. Again, since the certain event Ω has area 1, the probability of M is equal to the area of M, i.e., $P(M) = 1 - \left(\frac{2}{3}\right)^2 = \frac{5}{9}$. $\square$

Geometric Probability

Example

(**Buffon's needle problem**) Given a needle of length l dropped on a plane ruled with parallel lines d ($l < d$) units apart. What is the probability that the needle will cross a line?

Geometric Probability

Example

(**Buffon's needle problem**) Given a needle of length l dropped on a plane ruled with parallel lines d ($l < d$) units apart. What is the probability that the needle will cross a line?

Solution.

- Let A be the event that the needle will cross a line.
- Let x be the distance from the center of the needle to the closest line, see Fig. 1.3(b). Then x will be in the range $0 \leqslant x \leqslant d/2$.
- Let θ be the acute angle between the needle and the lines. Then θ will be in the range $0 \leqslant \theta \leqslant \pi$.

Geometric Probability

That is,

$$\Omega = \{(x, \theta) |\ 0 \leqslant x \leqslant d/2,\ 0 \leqslant \theta \leqslant \pi\}$$

in the $Ox\theta$ plane. Since the needle intersects with one line if and only if $x \leqslant \dfrac{l \sin\theta}{2}$, i.e.,

$$A = \{(x, \theta) |\ 0 \leqslant x \leqslant d/2,\ 0 \leqslant \theta \leqslant \pi, x \leqslant \frac{l \sin\theta}{2}\}.$$

So the probability required is

$$P(A) = \frac{L(A)}{L(\Omega)} = \frac{\int_0^\pi \frac{l \sin\theta}{2} \mathrm{d}\theta}{\frac{d\pi}{2}} = \frac{2l}{\pi d}.$$

Buffon's needle problem is a question first posed in 1777 by Georges-Louis Leclerc, Comte de Buffon. The solution can be used to design a Monte Carlo method for approximating the number π. $\square$

Geometric Probability

Is the definition of geometric probability perfect?

Geometric Probability

Is the definition of geometric probability perfect?

Let us see the following famous example.

Example

(**Bertrand paradox**) Consider an equilateral triangle inscribed in a circle. Suppose a chord of the circle is chosen at random. What is the probability that the chord is longer than a side of the triangle? Bertrand gave three arguments, all apparently valid, yet yielding different results.

Geometric Probability

(i) The "random midpoint" method: Choose a point anywhere within the circle and construct a chord with the chosen point as its midpoint. The chord is longer than a side of the inscribed triangle if the chosen point falls within a concentric circle of radius 1/2 the radius of the larger circle. The area of the smaller circle is one fourth the area of the larger circle, therefore the probability a random chord is longer than a side of the inscribed triangle is 1/4.

Geometric Probability

(ii) The "random endpoints" method: Choose two random points on the circumference of the circle and draw the chord joining them. To calculate the probability in question imagine the triangle rotated so its vertex coincides with one of the chord endpoints. Observe that if the other chord endpoint lies on the arc between the endpoints of the triangle side opposite the first point, the chord is longer than a side of the triangle. The length of the arc is one third of the circumference of the circle, therefore the probability that a random chord is longer than a side of the inscribed triangle is 1/3.

Geometric Probability

(ii) The "random radius" method: Choose a radius of the circle, choose a point on the radius and construct the chord through this point and perpendicular to the radius. To calculate the probability in question imagine the triangle rotated so a side is perpendicular to the radius. The chord is longer than a side of the triangle if the chosen point is nearer the center of the circle than the point where the side of the triangle intersects the radius. The side of the triangle bisects the radius, therefore the probability a random chord is longer than a side of the inscribed triangle is 1/2.

Section 1.3 Probabilities Defined on Events

- 1.3.1 Classical Probability
- 1.3.2 Geometric Probability
- 1.3.3 The Frequency Interpretation of Probability

The Frequency Interpretation of Probability

Suppose that some random procedure has several possible outcomes that are not necessarily equally likely.

How can we define the probability $P(A)$ of any eventuality A of interest?

For example, suppose that the procedure is the rolling of a die that is suspected to be weighted, or even clearly asymmetrical, in not being a perfect cube. What now is the probability of a six?

The Frequency Interpretation of Probability

What we usually do is to roll the die a large number n of times, and let $f(n)$ be the number of sixes shown. Then (provided the rolls were made under similar conditions) the symmetry between the rolls suggests that (at least approximately)

$$P(\text{six}) = \frac{f(n)}{n} = \frac{\text{number of sixes}}{\text{number of rolls}}$$

Furthermore, if you actually obtain an imperfect or weighted die and roll it many times, you will find that as n increases the ratio $f(n)/n$ always appears to be settling down around some asymptotic value. This provides further support for our taking $f(n)/n$ as an approximation to $P(\text{six})$.

The Frequency Interpretation of Probability

Let E be an random experiment, A be an random event. Suppose that E was repeated n times under similar conditions. Let $f_n(A)$ be the times that A occurs. The ration

$$F_n(A) = \frac{f_n(A)}{n}$$

is said to be the ***frequency*** of event A in the n trials. If n is large enough, the probability of event A will be approximated by $F_n(A)$.

The Frequency Interpretation of Probability

Theorem

For a random experiment E, *the frequency has the following properties:*

(i) *for every event* A, $F_n(A) \geqslant 0$,

(ii) $F_n(\Omega) = 1$,

(iii) *for every finite sequence of* n *disjoint events* $A_1, A_2, \cdots, A_n$,

$$F_n\Big(\bigcup_{i=1}^{n} A_i\Big) = \sum_{i=1}^{n} F_n(A_i)$$

The Frequency Interpretation of Probability

If we run Buffon's needle experiment a large number of times the proportion of crack crossings should be about the same as the probability of a crack crossing. More precisely, we will denote the number of crack crossings in the first n runs by f_n. In any case, if n is large, we should have

$$\pi \approx \frac{2ln}{f_n d} = \frac{2l}{F_n d}.$$

This is Buffon's famous estimate of π.

The frequency interpretation of probability is very important because the frequency can appropriately reflect the real probability and it is very simple to master.

Is this interpretation perfect?

The Frequency Interpretation of Probability

Is this interpretation perfect?

(i) there is no definite indication of an actual number that would be considered large enough;

(ii) this interpretation of probability rests on the important assumption that our process or experiment can be repeated many times under similar conditions. While the real-world experiment must not be completely controlled but must have some "random" features;

(iii) the probability of event A will be approximated by $F_n(A)$ do not mean $P(A)$ is the limit of $F_n(A)$;

The Frequency Interpretation of Probability

(iv) the frequency interpretation of probability is that it applies only to a problem in which there can be, at least in principle, a large number of similar repetition of a certain process. Many important problems are not of this type.

For example, the frequency interpretation of probability cannot be applied directly to the probability that a specific acquaintance will get married within the next two years or to the probability that a particular medical research project will lead to the development of a new treatment for a certain disease within a specified period of time.

Summary for Section 1.3

- **classical probability**:
  - E contains only different limited basic events,
  - all outcomes are equally likely to occur.
- **geometrical probability**:
  - the sample space is a measurable region,
  - the probability of every event $A \subset \Omega$ is proportional to the measure $L(A)$ and has nothing to do with its position and shape.
- **frequency**: do experiments

Each interpretation of probability, as we can see, has its appeal and its difficulties. We need a pure mathematical definition for probability for general cases.

Section 1.4 Probability Space

- 1.4.1 Axiomatic Definition of Probability
- 1.4.2 Properties of Probability

Axiomatic Definition of Probability

Definition

A collection $\mathscr{F}$ of subsets of Ω is a ***σ-algebra*** *if*
(i) $\Omega \in \mathscr{F}$;
(ii) $F \in \mathcal{F} \Longrightarrow \overline{F} \in \mathscr{F}$;
(iii) *if F_n is a countable collection of sets, $n = 1, 2, \cdots$ such that $F_n \in S$ for all n, then*

$$\bigcup_n F_n \in \mathscr{F}.$$

What is the intuition behind a σ-algebra?

If Ω represents the collection of possible outcomes of an experiment, a subset of Ω is called an event. Then, a σ-algebra represents the collection of all possible, interesting events from the viewpoint of a given experiment.

Axiomatic Definition of Probability

Example

Let the experiment be a coin toss (where we blow on the coin if it stands up straight!). Then $\Omega = \{H, T\}$. Let $\mathscr{F}_1 = \{\emptyset, \{H\}, \{T\}, \{H, T\}\}$ and $\mathscr{F}_2 = \{\emptyset, \{H, T\}\}$. It is easy to prove that both $\mathscr{F}_1$ and $\mathscr{F}_2$ are σ-algebra on Ω.

Now we know that the reason that Bertrand Paradox occurs is σ-algebra on Ω is not uniquely determined.

Axiomatic Definition of Probability

Definition

Let $P(A)(A \in \mathscr{F})$ be a non-negative set function on the σ-algebra $\mathscr{F}$. $P(A)$ is called the probability measure or ***probability of event*** *A if it satisfies the following three axioms:*

Axiom 1. *for every $A \in \mathscr{F}, P(A) \geqslant 0$;*

Axiom 2. $P(\Omega) = 1$;

Axiom 3. *(countable additivity) for every infinite sequence of countable disjoint events $A_1, A_2, \cdots$,*

$$P(\bigcup_{i=1}^{\infty} A_i) = \sum_{i=1}^{\infty} P(A_i).$$

The sets in σ-algebra $\mathscr{F}$ are called events. $\mathscr{F}$ is called to be the algebra of events. The triple $(\Omega, \mathscr{F}, P)$ is a ***probability space*** *or* ***probability triple****.*

Section 1.4 Probability Space

- 1.4.1 Axiomatic Definition of Probability
- 1.4.2 Properties of Probability

1.4.2 Properties of Probability

Theorem

$P(\emptyset) = 0.$

1.4.2 Properties of Probability

Theorem

$P(\emptyset) = 0.$

Proof. Consider the infinite sequence of events $A_1, A_2, \cdots$ such that $A_i = \emptyset$ for $i = 1, 2, \cdots$. In other words, each of the events in the sequence is just the empty set $\emptyset$. This sequence is a sequence of disjoint events, since $\emptyset \cap \emptyset = \emptyset$. Furthermore, $\cup_{i=1}^{\infty} A_i = \emptyset$. Therefore, it follows from Axiom 3 that

$$P(\emptyset) = P\Big(\bigcup_{i=1}^{\infty} A_i\Big) = \sum_{i=1}^{\infty} P(A_i) = \sum_{i=1}^{\infty} P(\emptyset).$$

This equation states that when the number $P(\emptyset)$ is added repeated in an infinite series, the sum of that series is simply the number $P(\emptyset)$. The only real number with this property is zero.

Properties of Probability

Theorem

For every finite sequence of countable disjoint events $A_1, A_2, \cdots, A_n$,

$$P(\bigcup_{i=1}^{n} A_i) = \sum_{i=1}^{n} P(A_i).$$

Properties of Probability

Theorem

For every finite sequence of countable disjoint events $A_1, A_2, \cdots, A_n$,

$$P\left(\bigcup_{i=1}^{n} A_i\right) = \sum_{i=1}^{n} P(A_i).$$

Proof. Consider the infinite sequence of events $A_1, A_2, \cdots$, in which $A_1, A_2, \cdots, A_n$ are the n given disjoint events and $A_i = \emptyset$ for $i > n$. Then the events in this infinite sequence are disjoint and $\cup_{i=1}^{\infty} A_i = \cup_{i=1}^{n} A_i$. Therefore,

$$\begin{aligned} P\left(\bigcup_{i=1}^{n} A_i\right) &= P\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} P(A_i) = \sum_{i=1}^{n} P(A_i) + \sum_{i=n+1}^{\infty} P(A_i) \\ &= \sum_{i=1}^{n} P(A_i) + 0 = \sum_{i=1}^{n} P(A_i) \end{aligned}$$

holds by Axiom 3. $\square$

Properties of Probability

Theorem

For every event A, $P(\overline{A}) = 1 - P(A)$.

Proof. Since A and $\overline{A}$ are disjoint events and $A \cup \overline{A} = \Omega$, $P(\Omega) = P(A) + P(\overline{A})$. By Axiom 2, we know $P(\Omega) = 1$, then $P(\overline{A}) = 1 - P(A)$. $\square$

Theorem

If $A \subset B$, then $P(B - A) = P(B) - P(A)$ and $P(A) \leqslant P(B)$.

Properties of Probability

Theorem

For every event A, $P(\overline{A}) = 1 - P(A)$.

Proof. Since A and $\overline{A}$ are disjoint events and $A \cup \overline{A} = \Omega$, $P(\Omega) = P(A) + P(\overline{A})$. By Axiom 2, we know $P(\Omega) = 1$, then $P(\overline{A}) = 1 - P(A)$. □

Theorem

If $A \subset B$, then $P(B - A) = P(B) - P(A)$ and $P(A) \leqslant P(B)$.

Proof. Since $A \subset B$, the event B can be treated as the union of the two disjoint events A and $B\overline{A}$. Therefore, $P(B) = P(A) + P(B\overline{A})$, i.e.,

$$P(B - A) = P(B\overline{A}) = P(B) - P(A).$$

Since $P(B - A) \geqslant 0$, $P(B) \geqslant P(A)$. □

Properties of Probability

Theorem

For every event A, $0 \leqslant P(A) \leqslant 1$.

Proof. It is known from Axiom 1 that $P(A) \geqslant 0$. Since $A \subset \Omega$ for every event A, Theorem 1.3.4 implies $P(A) \leqslant P(\Omega) = 1$, by Axiom 2. $\square$

Theorem

For every two events A and B,

$$P(A \cup B) = P(A) + P(B) - P(AB). \tag{3}$$

Properties of Probability

Proof. The event $A \cup B$ may be treated as the union of the three events $A\overline{B}, AB$ and $\overline{A}B$. Since the three events are disjoint,

$$P(A \cup B) = P(A\overline{B} \cup AB \cup \overline{A}B) = P(A\overline{B}) + P(AB) + P(\overline{A}B). \quad (4)$$

Furthermore, it can be seen that

$$P(A) = P(A\overline{B}) + P(AB)$$

and

$$P(B) = P(\overline{A}B) + P(AB).$$

We now use these last two equations to compute the quantity on the right-hand side of (3):

$$\begin{aligned} P(A) + P(B) - P(AB) &= P(A\overline{B}) + P(AB) + P(\overline{A}B) + P(AB) - P(AB) \\ &= P(A\overline{B}) + P(AB) + P(\overline{A}B), \end{aligned}$$

which is the right-hand side of (4). $\square$

Properties of Probability

Corralary

For every three events A_1, A_2 and A_3,

$$\begin{aligned} P(A_1 \cup A_2 \cup A_3) = & P(A_1) + P(A_2) + P(A_3) \\ & - [P(A_1A_2) + P(A_2A_3) + P(A_1A_3)] \\ & + P(A_1A_2A_3). \end{aligned} \tag{5}$$

Corralary

For every n events $A_1, A_2, \cdots, A_n$,

$$P(\bigcup_{i=1}^{n} A_i) = \sum_{i=1}^{n} P(A_i) - \sum_{i<j} P(A_iA_j) + \sum_{i<j<k} P(A_iA_jA_k) \tag{6}$$

$$- \textstyle\sum_{i<j<k<l} P(A_iA_jA_kA_l) + \cdots + (-1)^{n+1} P(A_1A_2 \cdots A_n).$$

Properties of Probability

Theorem

(i) *Let* $A_1, A_2, \cdots$ *be an infinite sequence of events such that* $A_1 \subset A_2 \subset \cdots$. *Then*

$$P\Big(\bigcup_{i=1}^{\infty} A_i\Big) = \lim_{n\to\infty} P(A_n). \tag{7}$$

(ii) *Let* $A_1, A_2, \cdots$ *be an infinite sequence of events such that* $A_1 \supset A_2 \supset \cdots$. *Then*

$$P\Big(\bigcap_{i=1}^{\infty} A_i\Big) = \lim_{n\to\infty} P(A_n). \tag{8}$$

Properties of Probability

Proof. (i) Let $A = \bigcup_{i=1}^{\infty} A_i$. Since $A_n \subset A_{n+1}$, we rewrite A_n by n disjoint events $A_1, A_2\overline{A_1}, \cdots, A_n\overline{A_{n-1}}$:

$$A_n = A_1 \cup A_2\overline{A_1} \cup \cdots \cup A_n\overline{A_{n-1}}.$$

Let $B_1 = A_1$, $B_m = A_m\overline{A_{m-1}}$, $m = 2, 3, \cdots$. Then

$$P(A) = P(\bigcup_{m=1}^{\infty} B_m) = \sum_{m=1}^{\infty} P(B_m) = \lim_{n\to\infty} \sum_{m=1}^{n} P(B_m).$$

Thus $\sum_{m=1}^{n} P(B_m) = P(\bigcup_{m=1}^{n} B_m) = P(A_n)$. So $P(\bigcup_{i=1}^{\infty} A_i) = \lim_{n\to\infty} P(A_n)$.

(ii) It can be obtained by defining $B_i = \overline{A_i}$. $\square$

Exercises

Example

Assume A and B are two events such that $P(A) = 1/3$ and $P(B) = 1/2$. Determine the value of $P(B - A)$ satisfying each of the following conditions. (a) A and B are disjoint; (b) $A \subset B$; (c) $P(AB) = 1/8$.

Solution. Since

$$P(B) = P(AB) + P(\overline{A}B) = P(AB) + P(B - A),$$

That is,

$$P(B - A) = P(B) - P(AB). \tag{9}$$

(a) Since A and B are disjoint means $AB = \emptyset$, we know

$$P(B - A) = P(B) - P(AB) = P(B) = 1/2.$$

(b) If $A \subset B$, then $AB = A$. Thus
$P(B - A) = P(B) - P(AB) = P(B) - P(A) = 1/6.$
(c) $P(B - A) = P(B) - P(AB) = 1/2 - 1/8 = 3/8.$ $\square$

Exercises

Example

Assume A and B are two events such that $P(B) = b$ and $P(A \cup B) = c$. Determine the value of $P(A\overline{B})$.

Solution. By equations (3) and (9),

$$\begin{aligned} P(A \cup B) =& P(A) + P(B) - P(AB) \\ =& P(B) + [P(A) - P(AB)] \\ =& P(B) + P(A\overline{B}). \end{aligned}$$

Thus $P(A\overline{B}) = P(A \cup B) - P(B) = c - b$.

$\square$

Exercises

Example

Select an integer from 1 to 1000 at random. Calculate the probability of the event that the integer is not divisible by 6 and 8.

Solution. Let A be the event that the integer is divisible by 6, and B be the event that the integer is divisible by 8. The required probability is

$$\begin{aligned} P(\overline{A}\ \overline{B}) =& P(\overline{A \cup B}) = 1 - P(A \cup B) \\ =& 1 - [P(A) + P(B) - P(AB)] \\ =& 1 - 166/1000 - 125/1000 + 41/1000 \approx 0.75. \end{aligned}$$

$\square$

Exercises

Example

(**The matching problem**) The matching experiment is a random experiment that can be formulated in a number of colorful ways.

An absent-minded secretary prepares n letters and envelopes to send to n different people, but then randomly stuffs the letters into the envelopes. A match occurs if a letter is inserted in the proper envelope.

n people with hats have had a bit too much to drink at a party. As they leave the party, each person randomly grabs a hat. A match occurs if a person gets his or her own hat.

Here we shall determine the probability of the event that at least one match occurs.

Exercises

Solution. Let A_i be the event that letter i is placed in the correct envelop ($i = 1, 2, \cdots, n$) (or a person gets his or her own hat) and we shall determine the value of $P(\bigcup_{i=1}^{n} A_i)$ by using equation (6). Since

$$P(A_i) = \frac{(n-1)!}{n!} = \frac{1}{n}, \qquad i = 1, 2, \cdots;$$

$$P(A_i A_j) = \frac{(n-2)!}{n!} = \frac{1}{n(n-1)}, \qquad 1 \leqslant i < j \leqslant n;$$

$$P(A_i A_j A_k) = \frac{(n-3)!}{n!} = \frac{1}{n(n-1)(n-2)}, \qquad 1 \leqslant i < j < k \leqslant n;$$

$$\vdots$$

$$P(A_1 \cdots A_n) = \frac{1}{n!},$$

$\square$

Exercises

Solution. we have

$$P\left(\bigcup_{i=1}^{n} A_i\right) = \binom{n}{1} \cdot \frac{1}{n} - \binom{n}{2} \cdot \frac{1}{n(n-1)} + \cdots + (-1)^{n+1} \frac{1}{n!}$$
$$= 1 - \frac{1}{2!} + \frac{1}{3!} + \cdots + (-1)^{n+1} \frac{1}{n!}.$$

$\square$

Exercises

This probability has the following interesting features. As $n \to \infty$, the value of $P(\bigcup_{i=1}^{n} A_i)$ approaches the following limit:

$$1 - \frac{1}{2!} + \frac{1}{3!} - \frac{1}{4!} + \cdots .$$

It is shown in books on elementary calculus that the sum of the infinite series on the right side of this equation is $1 - (1/e)$, where $e = 2.71828 \cdots$. Hence, $1 - (1 - 1/e) = 0.63212 \cdots$. It follows that for large value of n, the probability that at least one letter will be placed in the correct envelope is approximately 0.63212.

Exercises

The value of $P(\bigcup_{i=1}^{n} A_i)$ converges to the limit very rapidly. In fact, for $n = 7$ the exact value $P(\bigcup_{i=1}^{7} A_i)$ and the limiting value 0.63212 agree to four decimal places. Hence, regardless of whether seven letters are placed at random in seven or seven million letters are placed at random in seven million envelopes, the probability that at least one letter will be replaced in the correct envelope is 0.63212.

Summary for Section 1.4

- **Definition**
  - **Axiom 1**. for every $A \in \mathscr{F}, P(A) \geqslant 0$;
  - **Axiom 2**. $P(\Omega) = 1$;
  - **Axiom 3**. (countable additivity)
- **Properties**
  - $P(\emptyset) = 0$.
  - For disjoint events $A_1, A_2, \cdots, A_n$,

$$P\Big(\bigcup_{i=1}^{n} A_i\Big) = \sum_{i=1}^{n} P(A_i).$$

  - For every event A, $P(\overline{A}) = 1 - P(A)$.
  - If $A \subset B$, then $P(B - A) = P(B) - P(A)$ and $P(A) \leqslant P(B)$.
  - For every event A, $0 \leqslant P(A) \leqslant 1$.
  - For every two events A and B,

$$P(A \cup B) = P(A) + P(B) - P(AB).$$

Section 1.5 Conditional Probability

- 1.5.1 The Definition of Conditional Probability
- 1.5.2 The Multiplication Rule
- 1.5.3 Total Probability Formula
- 1.5.4 Bayes' Theorem

The Definition of Conditional Probability

The Monty Hall problem
Suppose that you are on a game show, and you are given the choice of three doors: Behind one door is a car; behind the others, goats. You pick a door, say No. 1, and the host, who knows what is behind the doors, opens another door, say No. 3, which has a goat. You are then given the option to switch your selection from Door 1 to the unopened Door 2. What is the probability that you will win the car if you switch your door selection to Door 2? Also, compute the probability that you will win the car if you do not switch.

What would you do?

The Definition of Conditional Probability

The critical factor in this problem is that the host knows what is behind the doors in advance. The probability of the event would be changed after we could know some information. That is Conditional Probabilities.

Let us consider a easier problem.

Suppose that we flip two fair coins and that each of the 4 possible outcomes is equally likely to occur and hence has the probability 1/4. Suppose that we observe that a head appears on the flip of one coin. Then given this information, what is the probability that (H, H) occurs? To calculate this probability we reason as follows:

The Definition of Conditional Probability

- Given that a head appears, it follows that there can be three outcomes $(H, H), (H, T)$ and (T, H).
- Since each of these outcomes originally had the same probability of occurring, they should still have equal probabilities.

That is, given that a head appears on the flip of one coin, then the (conditional) probability of each of the outcomes $(H, H), (H, T), (T, H)$ is $1/3$ while the (conditional) probability of the outcome (T, T) in the sample space is 0. Hence, the desired probability will be $1/3$.

The Definition of Conditional Probability

In fact, this probability can be calculated as follows.

Let $A = \{(H, H)\}$ and $B = \{$ a head appears on the flip of one coin$\}$.

$$\begin{aligned} P(A|B) = \frac{1}{3} &= \frac{\#\{(H,H)\}}{\#\{(H,H),(H,T),(T,H)\}} \\ &= \frac{\#AB}{\#B} = \frac{\frac{\#AB}{\#S}}{\frac{\#B}{\#S}} = \frac{P(AB)}{P(B)}. \end{aligned}$$

The Definition of Conditional Probability

Definition

Given two events A and B with $P(B) > 0$, the ***conditional probability of*** *A* ***given*** *B is defined as the quotient of the joint probability of A and B, and the probability of B:*

$$P(A|B) = \frac{P(AB)}{P(B)} \tag{11}$$

Theorem

If $P(B) > 0$, then the conditional probability $P(A|B)$ is also a probability, that is,
(i) *for every event $A, P(A|B) \geqslant 0$;*
(ii) $P(\Omega|B) = 1$;

The Definition of Conditional Probability

Theorem

(iii) *for every infinite sequence of countable disjoint events* $A_1, A_2, \cdots$,

$$P\Big(\bigcup_{i=1}^{\infty} A_i \Big| B\Big) = \sum_{i=1}^{\infty} P(A_i|B).$$

Proof. (i) Since $P(B) \geqslant P(AB)$ and $P(B) > 0$, we get $P(A|B) \geqslant 0$ by equation (11).

(ii) $P(\Omega|B) = \frac{P(\Omega B)}{P(B)} = \frac{P(B)}{P(B)} = 1.$

(iii) $P\Big(\bigcup\limits_{i=1}^{\infty} A_i \Big| B\Big) = \Big(P\big(\bigcup\limits_{i=1}^{\infty} A_i B\big)\Big) \Big/ P(B) = \sum\limits_{i=1}^{\infty} \frac{P(A_i B)}{P(B)} = \sum\limits_{i=1}^{\infty} P(A_i|B).$ $\square$

The Definition of Conditional Probability

Since the conditional probability is a probability, all properties of probabilities hold for conditional probabilities.

Property

If $P(B) > 0$, *then*
(i) $P(\emptyset|B) = 0$.
(ii) *For every finite sequence of countable disjoint events* $A_1, A_2, \cdots, A_n$,

$$P\Big(\bigcup_{i=1}^{n} A_i \Big| B\Big) = \sum_{i=1}^{n} P(A_i|B).$$

(iii) $P(\overline{A}|B) = 1 - P(A|B)$.
(iv) *If* $A \subset C$, *then* $P(C - A|B) = P(C|B) - P(A|B)$ *and* $P(A|B) \leqslant P(C|B)$.
(v) $P(A \cup C|B) = P(A|B) + P(C|B) - P(AC|B)$.

The Definition of Conditional Probability

Example

The probability that it is Friday and that a student is absent is 0.03. Since there are 5 school days in a week, the probability that is Friday is 0.2. What is the probability that a student is absent given that today is Friday?

Solution.

$$
\begin{aligned}
&P(\text{Absent}|\text{Friday}) \\
=&\frac{P(\text{Friday and Absent})}{P(\text{Friday})} \\
=&\frac{0.03}{0.2} = 0.15 = 15\%.
\end{aligned}
$$

□

The Definition of Conditional Probability

Example

Your neighbor has 2 children. You learn that he has a son, Joe. What is the probability that Joe's sibling is a brother?

Solution. The obvious answer that Joe's sibling is equally likely to have been born male or female suggests that the probability the other child is a boy is 1/2.

This is not correct!

Consider the experiment of selecting a random family having two children and recording whether they are boys or girls. Then, the sample space is $\Omega = \{BB, BG, GB, GG\}$, where, e.g., outcome means that the first-born child is a boy and the second-born is a girl.

The Definition of Conditional Probability

Solution. Assuming boys and girls are equally likely to be born, the 4 elements of Ω are equally likely.

The event, C, that the neighbor has a son is the set $C = \{BB, BG, GB\}$.

The event, D, that the neighbor has two boys (i.e., Joe has a brother) is the set $D = \{BB\}$. The desired probability is

$$P(D|C) = \frac{P(DC)}{P(C)} = \frac{P(\{BB\})}{P(\{BB, BG, GB\})} = 1/3.$$

$\square$

The Definition of Conditional Probability

Example

A machine produces parts that are either good (90%), slightly defective (2%), or obviously defective (8%). Produced parts get passed through an automatic inspection machine, which is able to detect any part that is obviously defective and discard it. What is the quality of the parts that make it through the inspection machine and get shipped?

Solution. Let A (resp., B, C) be the event that a randomly chosen shipped part is good (resp., slightly defective, obviously defective).

We are told that $P(A) = 0.90, P(B) = 0.02$, and $P(C) = 0.08$.

The Definition of Conditional Probability

Solution. Let A (resp., B, C) be the event that a randomly chosen shipped part is good (resp., slightly defective, obviously defective).

We are told that $P(A) = 0.90, P(B) = 0.02$, and $P(C) = 0.08$.

We want to compute the probability that a part is good given that it passed the inspection machine (i.e., it is not obviously defective), which is

$$P(A|\overline{C}) = \frac{P(A\overline{C})}{P(\overline{C})} = \frac{P(A)}{1 - P(C)} = \frac{0.90}{1 - 0.08} = 90/92 = 0.978.$$

$\square$

The Definition of Conditional Probability

Example

Ten fair dice are rolled at one time. What is the conditional probability of the event at least two land on 1 given the event at least one of the dice lands on 1.

Solution. Let A be the event that at least one of the dice lands on 1, B be the event that at least two land on 1, and C be the event that exactly one of the dice lands on 1.
We see that $B \subset A, C \subset A$ and $B = A - C$. By using the definition of classical probability,

$$P(A) = 1 - P(\overline{A}) = 1 - \frac{5^{10}}{6^{10}}, \qquad P(C) = \frac{10 \times 5^9}{6^{10}}.$$

The Definition of Conditional Probability

So the required probability is

$$\begin{aligned} P(B|A) &= \frac{P(AB)}{P(A)} = \frac{P(B)}{P(A)} = \frac{P(A) - P(C)}{P(A)} \\ &= \frac{1 - \dfrac{5^{10}}{6^{10}} - \dfrac{10 \times 5^9}{6^{10}}}{1 - \dfrac{5^{10}}{6^{10}}} \approx 0.615. \end{aligned}$$

$\square$

The Definition of Conditional Probability

Example

In describing the survival rate and life expectancy in a certain population, let A_N denote the event of a new-born to reach the age of N years. We are given that

$$P(A_{50}) = 0.913, \qquad P(A_{55}) = 0.881, \qquad P(A_{65}) = 0.746.$$

(a) What is the probability of a 50 years old man to reach the age of 55, i.e., what is $P(A_{55}|A_{50})$?
(b) If the probability that a man who just turned 65 will die within 5 years is 0.16, what is the probability for a man to survive till his 70th birthday, i.e., what is $P(A_{70})$?

The Definition of Conditional Probability

Solution. (a) Obviously, $A_{55} \cap A_{50} = A_{55}$. We have by definition,

$$P(A_{55}|A_{50}) = P(A_{55} \cap A_{50})/P(A_{50}) = P(A_{55})/P(A_{50}) \approx 0.965.$$

(b) Similarly, $P(A_{70}|A_{65}) = P(A_{70})/P(A_{65})$. So $P(A_{70}) = P(A_{65}) \cdot P(A_{70}|A_{65})$. We get

$$P(A_{70}|A_{65}) = 1 - 0.16 = 0.84.$$

Therefore,
$P(A_{70}) = P(A_{65}) \cdot P(A_{70}|A_{65}) = 0.746 \cdot 0.84 \approx 0.627.$ $\square$

Section 1.5 Conditional Probability

- 1.5.1 The Definition of Conditional Probability
- 1.5.2 The Multiplication Rule
- 1.5.3 Total Probability Formula
- 1.5.4 Bayes' Theorem

The Multiplication Rule

The additivity of probability in Section 1.4 helped us to calculate the probabilities of the events that at least one of the events occurs.

How to deal with the probability of the events that many events occur at one time?

The multiplication rule (also known as the "Law of Multiplication").

In a sense, the multiplication rule could be regarded as a perfect mathematical reflection of the Chinese phrase "proceed in an orderly way and step by step".

The Multiplication Rule

Theorem

Assume that $P(B) > 0$. *Then*

$$P(AB) = P(B) \cdot P(A|B). \tag{12}$$

Or if $P(A) > 0$, *then*

$$P(AB) = P(A) \cdot P(B|A). \tag{13}$$

Theorem

Suppose that $A_1, A_2, \cdots, A_n$ *are events satisfying* $P(A_1 A_2 \cdots A_{n-1}) > 0$. *Then*

$$\begin{aligned} & P(A_1 A_2 \cdots A_n) \\ = & P(A_1) P(A_2|A_1) P(A_3|A_1 A_2) \cdots P(A_n|A_1 A_2 \cdots A_{n-1}). \end{aligned}$$

The Multiplication Rule

Example

Suppose that five good fuses and two defective ones have been mixed up. To find the defective fuses, we test them one-by-one, at random and without replacement. What is the probability that we find both of the defective fuses in the first two tests?

Solution. Let D_1 (resp., D_2) be the event that we find a defective fuse in the first (resp., second) test. Then

$$P(D_1 D_2) = P(D_1)P(D_2|D_1) = \frac{2}{7}\frac{1}{6} = \frac{1}{21}.$$

$\square$

The Multiplication Rule

Example

If six cards are selected at random (without replacement) from a standard deck of 52 cards, what is the probability there will be no pairs? (two cards of the same denomination)

Solution. Let A_i be the event that the first i cards have no pair among them, $i = 1, 2, \cdots, 6$.
Of course, one can solve the problem directly using counting techniques: define the sample space to be (equally likely) ordered sequences of 6 cards; then, $\#\Omega = 52 \cdot 51 \cdot 50 \cdots 47$, and the event A_6 has $52 \cdot 48 \cdot 44 \cdots 32$ elements. So

$$P(A_6) = \frac{\#A}{\#\Omega} = \frac{52 \cdot 48 \cdot 44 \cdots 32}{52 \cdot 51 \cdot 50 \cdots 47}.$$

The Multiplication Rule

Solution. Here we use the multiplication rule. We want to compute $P(A_6)$, which is actually the same as $P(A_1A_2\cdots A_6)$, since $A_6 \subset A_5 \cdots \subset A_1$, implying that $A_1A_2\cdots A_6 = A_6$. We get

$$\begin{aligned} P(A_1A_2\cdots A_6) =& P(A_1)\cdot\frac{P(A_1A_2)}{P(A_1)}\cdot\cdots\cdot\frac{P(A_1A_2\cdots A_6)}{P(A_1A_2\cdots A_5)} \\ =& \frac{52}{52}\frac{48}{51}\frac{44}{50}\frac{40}{49}\frac{36}{48}\frac{32}{47}. \end{aligned}$$

□

The Multiplication Rule

Example

(**Polya urn mode**) Suppose that an urn contains r red balls and b blue balls ($r \geqslant 2, b \geqslant 2$). Suppose that one ball is drawn randomly from the urn and its color is observed; it is then replaced in the urn, and c additional balls of the same color is added to the urn, and the selection process is repeated four times. We shall determine the probability of obtaining the sequence of outcomes red, blue, red, blue.

Solution. Let R_j denote the event that a red ball is obtained on the jth draw and B_j the event that a blue ball is obtained on the jth draw ($j = 1, 2, 3, 4$). Then

$$\begin{aligned} P(R_1B_2R_3B_4) &= P(R_1)P(B_2|R_1)P(R_3|R_1B_2)P(B_4|R_1B_2R_3) \\ &= \frac{r}{r+b} \cdot \frac{b}{r+b+c} \cdot \frac{r+c}{r+b+2c} \cdot \frac{b+c}{r+b+3c}. \end{aligned}$$

$\square$

The Multiplication Rule

Theorem

Suppose that $A_1, A_2, \cdots, A_n, B$ *are events such that* $P(A_1A_2\cdots A_{n-1}|B) > 0$. *Then*

$$P(A_1A_2\cdots A_n|B) = P(A_1|B)P(A_2|A_1B)P(A_3|A_1A_2B)\cdots P(A_n|A_1A_2\cdots A_{n-1}B).$$

Theoretically speaking, the multiplication rule holds no matter which one of these events happens first. But in actual applications, we usually use the multiplication rule in the situation that the events occur successively.

Total Probability Formula

Assume that A is the event caused by $B_1, B_2, \cdots, B_n$. The "Law of Total Probability" allows us to compute the probability of an event A by conditioning on cases, according to a partition $B_1, B_2, \cdots, B_n$ of the sample space.

Definition

Let Ω denote the sample space of some experiment. n events $B_1, B_2, \cdots, B_n$ are said to form a ***partition*** *of Ω if these events satisfy:*

(i) $B_1, B_2, \cdots, B_n$ *are disjoint and*

(ii) $\displaystyle\bigcup_{i=1}^{n} B_i = \Omega$.

Section 1.5 Conditional Probability

- 1.5.1 The Definition of Conditional Probability
- 1.5.2 The Multiplication Rule
- 1.5.3 Total Probability Formula
- 1.5.4 Bayes' Theorem

Total Probability Formula

One way to partition Ω is to break it into sets B and $\overline{B}$, for any event B. Thus for event A, we have

$$\begin{aligned} P(A) &= P(A\Omega) = P(A \cap (B \cup \overline{B})) = P(AB \cup A\overline{B}) \\ &= P(AB) + P(A\overline{B}) = P(B) \cdot P(A|B) + P(\overline{B}) \cdot P(A|\overline{B}) \end{aligned}$$

It is the simplest form of the law of total probability. More generally,

Theorem

Suppose that the events $B_1, B_2, \cdots, B_n$ form a partition of the sample space Ω and $P(B_i) > 0$ for $i = 1, 2, \cdots, n$. Then, for every event A in Ω,

$$P(A) = \sum_{i=1}^{n} P(B_i)P(A|B_i). \tag{14}$$

Total Probability Formula

Example

Two cards from an ordinary deck of 52 cards are missing. What is the probability that a random card drawn from this deck is a spade?

Solution. Let A be the event that the randomly drawn card is a spade. Let B_i be the event that i spades are missing from the 50-card (defective) deck, for $i = 0, 1, 2$.
By conditioning on how many spades are missing from the original (good) deck, we get

$$\begin{aligned} P(A) &= P(B_0)P(A|B_0) + P(B_1)P(A|B_1) + P(B_2)P(A|B_2) \\ &= \frac{13}{50}\frac{\binom{13}{0}\binom{39}{2}}{\binom{52}{2}} + \frac{12}{50}\frac{\binom{13}{1}\binom{39}{1}}{\binom{52}{2}} + \frac{11}{50}\frac{\binom{13}{2}\binom{39}{0}}{\binom{52}{2}} \approx \frac{1}{4}. \end{aligned}$$

$\square$

Total Probability Formula

Example

Your neighbor has 2 children. He picks one of them at random and comes by your house; he brings a boy named Joe (his son). What is the probability that Joe's sibling is a brother?

Solution. There is a mechanism by which the son was selected that gave you the information that your neighbor has a boy (the mechanism was random selection).
Let C be the event that your neighbor has a son隆卤and C' the event that your neighbor randomly chose one of his 2 children, and that chosen child is a son.
We note that $C' \subseteq C$ since event C' happening implies that event C happens (if he chooses a son at random, you know for sure that he has a son!). It does not go the other way, though: C' does not imply C (just because he has a son does not mean that he chose that son at random).

Total Probability Formula

Solution. We want to compute $P(D|C') = \dfrac{P(DC')}{P(C')}$, where $P(DC') = P(\{BB\}) = 1/4$ and

$$\begin{aligned} P(C') &= P(C'|\{BB\})P(\{BB\}) + P(C'|\{BG\})P(\{BG\}) \\ &\quad + P(C'|\{GB\})P(\{GB\}) + P(C'|\{GG\})P(\{GG\}) \\ &= 1 \cdot \frac{1}{4} + \frac{1}{2} \cdot \frac{1}{4} + \frac{1}{2} \cdot \frac{1}{4} + 0 \cdot \frac{1}{4} = 1/2. \end{aligned}$$

So $P(D|C') = \dfrac{P(DC')}{P(C')} = \dfrac{1/4}{1/2} = 1/2.$ $\square$

Total Probability Formula

How to ask a sensitive question?

- What proportion of people use illegal drugs?
- How many students ever cheated on an exam?
- ...

Surveying people directly and asking these types of sensitive questions are not likely to get honest responses and useful data.

Using probabilistic methods, statisticians have developed interesting ways, total probability methods, to ask sensitive questions that protect confidentiality.

Total Probability Formula

Example

Respondents are given a coin and told to flip it in private, not letting anyone see the outcome. If it lands heads, they answer the sensitive question of interest (e.g. "Have you ever taken illegal drugs?"). If tails, they answer an innocuous question such as "Were you born in the first half of the year?" The respondent reports a yes or no, but does not say which question they actually answered. From a sample of such yes-no responses, how can statisticians estimate the parameter of interest, such as the proportion of people who have ever taken illegal drugs?

Total Probability Formula

Solution. Let Y and N denote responses of yes and no, respectively. Let A denote the sensitive question and B the innocuous question. The unknown parameter that surveyors want to estimate is $p = P(Y|A)$, the probability that someone answers yes given that they were asked the sensitive question. We assume that the innocuous question is (i) easy to answer and (ii) has a known probability of yes and no, in this case 0.5 each.

Consider the unconditional probability $P(Y)$. By the law of total probability,

$$\begin{aligned} P(Y) &= P(Y|A)P(A) + P(Y|B)P(B) \\ &= p \cdot \frac{1}{2} + \frac{1}{2} \cdot \frac{1}{2} \\ &= \frac{p}{2} + \frac{1}{4}. \end{aligned}$$

Total Probability Formula

Solution. When this survey is given to n people, the finial data will consist of n yes's and no's. The proportion of yes's is a simulated estimate of the unknown $P(Y)$. And thus

$$\frac{p}{2} + \frac{1}{4} = P(Y) \approx \frac{\text{Numbers of yes's in the sample}}{n}.$$

Solving for p gives

$$p \approx 2\left(\frac{\text{Numbers of yes's in the sample}}{n} - \frac{1}{4}\right),$$

which is the final estimate of the parameter of interest. $\square$

Section 1.5 Conditional Probability

- 1.5.1 The Definition of Conditional Probability
- 1.5.2 The Multiplication Rule
- 1.5.3 Total Probability Formula
- 1.5.4 Bayes' Theorem

Bayes' Theorem

Suppose that someone told you they had a nice conversation with someone on the train. Not knowing anything else about this conversation, the probability that they were speaking to a woman is 50%. Now suppose that they also told you that this person had long hair. It is now more likely they were speaking to a woman, since women are more likely to have long hair than men. The following result, which is known as *Bayes' theorem*, can be used to calculate the probability that the person is a woman.

Bayes' Theorem

Theorem

(***Bayes' Theorem***) *Let the events* $B_1, B_2, \cdots, B_n$ *form a partition of the sample space* Ω *such that* $P(B_i) > 0$ *for* $i = 1, 2, \cdots, n$, *and let* A *be an event such that* $P(A) > 0$. *Then, for* $i = 1, 2, \cdots, n$,

$$P(B_i|A) = \frac{P(B_i)P(A|B_i)}{\sum_{j=1}^{n} P(B_j)P(A|B_j)} \tag{15}$$

Proof. By the definition of conditional probability, $P(B_i|A) = \dfrac{P(B_iA)}{P(A)}$. By the multiplication rule for conditional probabilities, equation (12), $P(B_iA) = P(B_i)P(A|B_i)$. Thus $P(A) = \sum_{j=1}^{n} P(B_j)P(A|B_j)$. The equation 15 holds. □

Bayes' Theorem

Solution to The Monty Hall Problem. Suppose that $\Omega = \{1, 2, 3\}$, where outcome i means that the car is behind door i. Let $A = \{$ Host open Door 3 $\} = \{1, 2\}$ and $B = \{$ You choose Door 1$\}$. The probability that you win by switching to Door 2 given that he tells you Door 3 has a goat is:

$$
\begin{aligned}
& P(\{2\}|AB) \\
= & \frac{P(AB \cap \{2\})}{P(AB)} = \frac{P(AB|\{2\})P(\{2\})}{P(AB)} \\
= & \frac{P(AB|\{2\})P(\{2\})}{P(AB|\{1\})P(\{1\}) + P(AB|\{2\})P(\{2\}) + P(AB|\{3\})P(\{3\})} \\
= & \frac{1 \cdot \frac{1}{3}}{\frac{1}{2} \cdot \frac{1}{3} + 1 \cdot \frac{1}{3} + 0 \cdot \frac{1}{3}} = \frac{2}{3}.
\end{aligned}
$$

While $P(\{1\}|AB) = 1 - P(\{2\}|AB) = 1/3$. $\square$

Bayes' Theorem

Example

A new test has been devised for detecting a particular type of cancer. If the test is applied to a person who has this type of cancer, the probability that the person will have a positive reaction is 0.95 and the probability that the person will have a negative reaction is 0.05. If the test is applied to a person who does not have this type of cancer, the probability that the person will have a positive reaction is 0.05 and the probability that the person will have a negative reaction is 0.95. Suppose that in the general population, one person out of every 100,000 people has this type of cancer. If a person selected at random has a positive reaction to the test, what is the probability that he has this type of cancer?

Bayes' Theorem

Solution. Let A denote the event that a person has a positive reaction to the test, B_1 denote the event that he has this type of cancer and B_2 denote the event that he does not have this type of cancer. Now we have known that $P(A|B_1) = 0.95, P(\overline{A}|B_1) = 0.05, P(A|B_2) = 0.05, P(\overline{A}|B_2) = 0.95$ and $P(B_1) = 1/100000$. It now follows from Bayes' theorem that

$$\begin{aligned} P(B_1|A) &= \frac{P(B_1)P(A|B_1)}{P(B_1)P(A|B_1) + P(B_2)P(A|B_2)} \\ &= \frac{(1/100000)\cdot(0.95)}{(1/100000)\cdot(0.95) + (99999/100000)\cdot(0.05)} \approx 0.00019. \end{aligned}$$

$\square$

Bayes' Theorem

It seems impossible! The result is contrary to our common sense. So in practical applications, doctors generally increased the probability $P(B_1)$ through some other medical examinations in advance to increase the accuracy of the test. For example, if $P(B_1) = 0.2$, then $P(B_1|A) \approx 0.83$. This example illustrates not only the use of Bayes' theorem, but also the importance of taking into account all of the information available in a problem.

Bayes' Theorem

Example

Three different machines M_1, M_2 and M_3 were used for producing a large batch of similar manufactured items. Suppose that 20 percent of the items were produced by machine M_1, 30 percent of the items were produced by machine M_2, and 50 percent of the items were produced by machine M_3. Suppose further that 1 percent of the items produced by machine M_1 are defective, that 2 percent of the items produced by machine M_2 are defective, and that 3 percent of the items produced by machine M_3 are defective. Finally, suppose that one item is selected at random from the entire batch, and it is found to be defective. We shall determine the probability that this item was produced by machine $M_i (i = 1, 2, 3)$.

Bayes' Theorem

Solution. Let B_i be the event that the selected item was produced by machine $M_i (i = 1, 2, 3)$, and let A be the event that the selected item is defective. We must evaluate the conditinal probability $P(B_2|A)$. The probability $P(B_i)(i = 1, 2, 3)$ is as follows:

$$P(B_1) = 0.2, \qquad P(B_2) = 0.3, \qquad P(B_3) = 0.5.$$

Furthermore, the probability $P(A|B_i)$ that an item produced by machine M_i will be defective is:
$P(A|B_1) = 0.01, P(A|B_2) = 0.02, P(A|B_3) = 0.03$. It now follows from Bayes' theorem that

$$\begin{aligned} P(B_1|A) &= \frac{P(B_1)P(A|B_1)}{P(B_1)P(A|B_1) + P(B_2)P(A|B_2) + P(B_3)P(A|B_3)} \\ &= \frac{(0.2)(0.1)}{(0.2)(0.1) + (0.3)(0.02) + (0.5)(0.03)} = 0.087. \end{aligned}$$

By the similar way, we obtain
$P(B_2|A) = 0.261, P(B_3|A) = 0.652.$ □

Bayes' Theorem

Remark

(i) *A probability like* $P(B_i)$ *is called the* ***prior probability*** *that the selected item will have been produced by machine* M_i*, because* $P(B_i)$ *is the probability of this event before the item is selected and before it is known whether the selected item is defective or nondefective.*

A probability like $P(B_i|A)$ *is called the* ***posterior probability*** *that the selected item was produced by machine* M_i*, because it is the probability of this event after it is known that the selected item is defective.*

Bayes' Theorem

Remark

(ii) *From above example, we can observe that*

$$P(B_1) = 0.2 > P(B_1|A) = 0.087,$$

$$P(B_3) = 0.5 < P(B_3|A) = 0.652.$$

That means there is no stationary numerical relationship between prior probability and posterior probability.

Bayes' Theorem

Example

(**Witness reliability**) After a robbery the thief jumped into a taxi and disappeared. An eyewitness on the crime scene is telling the police that the cab is yellow. Now we have known the following information:
(i) In that particular city 80% of taxis are black and 20% of taxis are yellow.
(ii) Eyewitness are not always reliable and from past experience it is expected that an eyewitness is 80% accurate. He will identify the color of a taxi accurately (yellow or black) 8 out 10 times.

Equipped with this information, can we make sure eyewitness' testimony is worth something?

Bayes' Theorem

Solution. Let
$\{true = Y\}$ be the event that the color of the taxi was actually yellow and
$\{true = B\}$ the event that it was black. Let
$\{report = Y\}$ be the event that the eyewitness identified the color of the taxi as actually yellow and
$\{report = B\}$ the event that it was reported as black.

The goal is to compute $P(true = Y | report = Y)$. By using Bayes' formula,

$$\begin{aligned} & P(true = Y | report = Y) \\ = & \frac{P(report = Y | true = Y) P(true = Y)}{P(report = Y)}, \end{aligned}$$

Bayes' Theorem

Solution. where $P(true = Y) = 0.2$ (this is the prior probability) and $P(report = Y|true = Y) = 0.8$ (this is the accuracy of witness testimony). Next to compute $P(report = Y)$, we need the actual color of the taxi. By using the law of total probability,

$$\begin{aligned} P(report = Y) =& P(report = Y|true = Y)P(true = Y) \\ &+ P(report = Y|true = B)P(true = B) \\ =& 0.8 \times 0.2 + 0.2 \times 0.8 = 0.32. \end{aligned}$$

Putting everything together we find

$$\begin{aligned} P(true = Y|report = Y) =& \frac{P(report = Y|true = Y)P(true = Y)}{P(report = Y)} \\ =& \frac{0.16}{0.32} = \frac{1}{2} \end{aligned}$$

and so the eyewitness' testimony does not provide much useful certainty. □

Bayesians VS Frequentists

Bayesians

(i) Probability is subjective and can be applied to single events based on degree of confidence or beliefs. For example, Bayesian can refer to tomorrow's weather as having 50% of rain, whereas this would not make sense to a Frequentist because tomorrow is just one unique event, and cannot be referred to as a relative frequency in a large number of trials.

(ii) Parameters are random variables that has a given distribution, and other probability statements can be made about them.

(iii) Probability has a distribution over the parameters, and point estimates are usually done by either taking the mode or the mean of the distribution.

Bayes' Theorem

Frequentists

(i) Probability is objective and refers to the limit of an event's relative frequency in a large number of trials. For example, a coin with a 50% probability of heads will turn up heads 50% of the time.

(ii) Parameters are all fixed and unknown constants.

(iii) Any statistical process only has interpretations based on limited frequencies. For example, a 95% C.I. of a given parameter will contain the true value of the parameter 95% of the time.

Summary for Section 1.5

- **Conditional Probability**
  $P(B|A) = P(AB)/P(A)$.
- **The multiplicational rule**
  $P(AB) = P(A)P(B|A)$.
- **Total probability formula**
  $P(A) = \sum_{i=1}^{n} P(B_i)P(A|B_i)$.
- **Bayes' Theorem**

$$P(B_i|A) = \frac{P(B_i)P(A|B_i)}{\sum_{j=1}^{n} P(B_j)P(A|B_j)}$$

where the events $B_1, B_2, \cdots, B_n$ form a partition of the sample space Ω such that $P(B_i) > 0$ for $i = 1, 2, \cdots, n$.

Section 1.6 Independent of Events

- 1.6.1 Independence of Two Events
- 1.6.2 Independence of Several Events
- 1.6.3 Bernoulli Trials

Independence of Two Events

Sometimes an event can affect the next event. We call those Dependent Events, because what happens depends on what happened before.

Consider the following examples:

(i) Landing on heads after tossing a coin AND rolling a 5 on a single 6-sided die.

(ii) Choosing a marble from a jar AND landing on heads after tossing a coin.

(iii) Choosing a 3 from a deck of cards, replacing it, AND then choosing an ace as the second card.

(iv) Rolling a 4 on a single 6-sided die, AND then rolling a 1 on a second roll of the die.

Independent Events are not affected by previous events.

This is an important idea!

Independence of Two Events

The "operational independence" described indicates that knowledge that one of the events has occurred does not affect the likelihood that the other will occur. For a pair of events A and B, this is the condition

$$P(A|B) = P(A) \text{ or } P(B|A) = P(B).$$

Occurrence of the event A is not 隆掳conditioned by隆卤occurrence of the event B. If the occurrence of B does not affect the likelihood of the occurrence of A, we should be inclined to think of the events A and B as being independent in a probability sense.

Independence of Two Events

We take our clue from the condition $P(A|B) = P(A)$ to define the independence of two events.

Definition

Two events A and B are ***independent*** *if*

$$P(AB) = P(A)P(B). \tag{16}$$

It is obvious that if $P(A) > 0$ and $P(B) > 0$, then independence is equivalent to the statement that the conditional probability of one event given the other is the same as the unconditional probability of the event:

$$P(A|B) = P(A) \Leftrightarrow P(B|A) = P(B) \Leftrightarrow P(AB) = P(A)P(B).$$

Independence of Two Events

The independence of the events in the examples at the beginning of this section are obvious. But things do not always like those.

Example

Let three fair coins be tossed.
Let $A = \{$ all heads or all tails $\}$, $B = \{$ at least two heads $\}$, and $C = \{$ at most two tails $\}$. Of the pairs of events, (A, B), (A, C), and (B, C), which are independent and which are dependent?

Solution. The events can be written explicitly:

$$\begin{aligned} A =& \{HHH, TTT\}, \\ B =& \{HHH, HHT, HTH, THH\}, \\ C =& \{HHH, HHT, HTH, THH, HTT, THT, TTH\}. \end{aligned}$$

Independence of Two Events

Solution. Then

$$
\begin{aligned}
P(AB) &= 1/8 = (2/8)(4/8) = P(A) \cdot P(B), \\
P(AC) &= 1/8 \neq (2/8)(7/8) = P(A) \cdot P(C), \\
P(BC) &= 4/8 \neq (4/8)(7/8) = P(B) \cdot P(C).
\end{aligned}
$$

Thus A and B are independent. A and C are dependent. B and C are dependent. $\square$

Independence of Two Events

Example

A plant gets two independent genes for flower color, one from each parent plant. If the genes are identical, then the flowers are uniformly of that color; if they are different, then the flowers are striped in those two colors. The genes for the colors pink, crimson, and red occur in the population in the proportions $p : q : r$, where $p + q + r = 1$. A given plant's parents are selected at random; let A be the event that its flowers are at least partly pink, and let B be the event that its flowers are striped.

(a) Find $P(A)$ and $P(B)$.

(b) Show that A and B are independent if $p = 2/3$ and $r = q = 1/6$.

(c) Are these the only values of p, q, and r such that A and B are independent?

Independence of Two Events

Solution. (a) With an obvious notation (P for pink, C for crimson, and R for red), we have

$$P(PP) = P(P)P(P) = p^2, \qquad \text{(by parents independence)},$$

because P occurs with probability p. Likewise,

$$P(PR) = P(R)P(P) = rp = P(RP).$$

Hence,

$$\begin{aligned} P(A) &= P(PP \cup PR \cup PR \cup PC \cup CP) \\ &= p^2 + 2pr + 2pq \\ &= 1 - (1-p)^2, \end{aligned}$$

because $p + q + r = 1$. (Can you see how to get this last expression directly?)
Similarly,

$$P(B) = P(PC \cup CP \cup PR \cup RP \cup RC \cup CR) = 2(pq + qr + rp).$$

Independence of Two Events

Solution. (b) The events A and B are independent, if and only if,

$$\begin{aligned} P(A)P(B) =& P(AB) = P(PC \cup CP \cup PR \cup RP) \\ =& 2(pq + pr). \end{aligned}$$

From part (a), this is equivalent to

$$(1 - (1-p)^2)(pq + qr + pr) = p(q + r), \tag{17}$$

and this is satisfied by the given values of p, q, and r.
(c) No. Rearranging equation (17), we see that A and B are independent for any values of q and r lying on the curve $rq = 2rq(q + r) + r^3 + q^3$, in the $r - q$ plane. You may care to amuse yourself by showing that this is a loop from the origin. Outside the loop, A and B are attractive; inside the loop, A and B are repellent. $\square$

Independence of Two Events

The terms independent and disjoint sound vaguely similar but they are actually very different.

Theorem

Suppose that A and B are disjoint events for an experiment, each with positive probability. Then A and B are dependent.

Proof. Since A and B are disjoint events, $P(AB) = P(\emptyset) = 0$. While $P(A)P(B) > 0$ because $P(A) > 0, P(B) > 0$. So $P(AB) \neq P(A)P(B)$. $\square$

Independence of Two Events

Theorem

If A and B are independent events in an experiment, then each of the following pairs of events is independent: (i) $\overline{A}$ *and* B*;* (ii) A *and* $\overline{B}$*;* (iii) $\overline{A}$ *and* $\overline{B}$*.*

Proof. We prove case (i). Suppose that A and B are independent. Then by the difference rule,

$$\begin{aligned} P(\overline{A}B) =& P(B) - P(AB) = P(B) - P(A)P(B) \\ =& [1 - P(A)]P(B) = P(\overline{A})P(B). \end{aligned}$$

Hence $\overline{A}$ and B are independent.
(ii) and (iii) can be obtained by the similar way. $\square$

Independence of Two Events

Example

Let A and B be independent events with $P(A) = 1/4$ and $P(A \cup B) = 2P(B) - P(A)$. Find the values of $P(B)$, $P(A|B)$ and $P(\overline{B}|A)$.

Solution. Since A and B are independent,

$$P(A \cup B) = P(A) + P(B) - P(AB) = P(A) + P(B) - P(A)P(B).$$

Thus,

$$1/4 + P(B) - (1/4)P(B) = 2P(B) - 1/4.$$

This implies that $P(B) = 2/5$.
Since A and B are independent,

$$P(A|B) = P(A) = 1/4 \quad \text{and} \quad P(\overline{B}|A) = P(\overline{B}) = \frac{3}{5}.$$

$\square$

Section 1.6 Independent of Events

- 1.6.1 Independence of Two Events
- 1.6.2 Independence of Several Events
- 1.6.3 Bernoulli Trials

Independence of Several Events

How to define the independence of three events A, B, C? How to express the occurrence of any event of the three events has nothing to do with the occurrence of the other two events?

Definition

Three events A, B and C in the sample space S of a random experiment are said to be ***mutually independent*** *(or independent) if*

$$\begin{aligned} P(AB) &= P(A)P(B), \\ P(AC) &= P(A)P(C), \\ P(BC) &= P(B)P(C), \\ P(ABC) &= P(A)P(B)P(C). \end{aligned}$$

Events A, B and C are said to be ***pairwise independent*** *if the first three equations hold.*

Independence of Several Events

Example

Suppose that a balanced coin is independently tossed two times. Define the following events:
A: Head appears on the first toss;
B: Head appears on the second toss;
C: Both tosses yield the same outcome.
Are events A, B and C mutually independent?

Solution. The sample space Ω consists of the following outcomes: $\Omega = \{HH, HT, TH, TT\}$ and the events listed above are $A = \{HH, HT\}$, $B = \{TH, HH\}$, $C = \{HH, TT\}$.
Since the coin is balanced, that is, all outcomes are assigned the same probability, namely $1/4$, we get

Independence of Several Events

Solution.

$$P(A) = P(B) = P(C) = 2/4 = 1/2,$$
$$P(AB) = P(AC) = P(BC) = \frac{1}{4},$$
$$P(ABC) = P(\{HH\}) = 1/4.$$

Therefore,

$$P(AB) = P(A)P(B), \quad P(AC) = P(A)P(C),$$

$$P(BC) = P(B)P(C).$$

However,

$$P(ABC) \neq P(A)P(B)P(C).$$

So A, B and C are pairwise independent. $\square$

Independence of Several Events

Example

Toss two different standard dice, white and black. Define the following events:
$A_1 = \{\text{first die} = 1, 2 \text{ or } 3\}$;
$A_2 = \{\text{first die} = 3, 4 \text{ or } 5\}$;
$A_3 = \{\text{sum of faces is } 9\}$.
Are events A_1, A_2 and A_3 mutually independent?

Solution. The sample space Ω of the outcomes consists of all ordered pairs $(i, j), i, j = 1, \cdots, 6$, i.e.,
$\Omega = \{(1,1), (1,2), ..., (6,6)\}$. Since the two dice are standard, each point in S has probability $1/36$. So the probability of the events are

$$P(A_1) = 1/2, \quad P(A_2) = 1/2, \quad P(A_3) = 1/9.$$

Independence of Several Events

Solution. Obviously,

$$\begin{aligned}
&A_1A_2 = \{(3,1),(3,2),(3,3),(3,4),(3,5),(3,6)\},\\
&A_1A_3 = \{(3,6)\},\ A_2A_3 = \{(3,6),(4,5),(5,4)\},\\
&A_1A_2A_3 = \{(3,6)\}.
\end{aligned}$$

It follows that

$$P(A_1A_2A_3) = 1/36 = (1/2)(1/2)(1/9) = P(A_1)P(A_2)P(A_3).$$

However,

$$\begin{aligned}
P(A_1A_2) &= 1/6 \neq 1/4 = P(A_1)P(A_2),\\
P(A_1A_3) &= 1/36 \neq 1/18 = P(A_1)P(A_3),\\
P(A_2A_3) &= 1/12 \neq 1/18 = P(A_2)P(A_3).
\end{aligned}$$

So events A_1, A_2 and A_3 are not mutually independent. $\square$

Independence of Several Events

Definition

Events $A_1, A_2, \cdots, A_n$ *are* ***pairwise independent*** *if every possible pair of these events are independent, i.e.,* $P(A_j A_k) = P(A_j) \cdot P(A_k)$ *for all* $j, k (j \neq k)$.

Definition

The n *events* $A_1, A_2, \cdots, A_n$ *are* ***independent*** *if, for every subset* $A_{i_1}, A_{i_2}, \cdots, A_{i_k}$ *of these events* $(k = 2, 3, \cdots, n)$,

$$P(A_{i_1} A_{i_2} \cdots A_{i_k}) = P(A_{i_1}) P(A_{i_2}) \cdots P(A_{i_k}). \tag{18}$$

Independence of Several Events

Theorem

The n events $A_1, A_2, \cdots, A_n$ are ***independent*** *if and only if for any subset $A_{i_1}, A_{i_2}, \cdots, A_{i_k}$,*

$$P(A_{i_1}^* A_{i_2}^* \cdots A_{i_k}^*) = P(A_{i_1}^*) P(A_{i_2}^*) \cdots P(A_{i_k}^*), \quad (k = 2, 3, \cdots, n) \tag{19}$$

where A_i^* denotes either A_i or $\overline{A}_i$ (the same on both sides of the equation).

Independence of Several Events

Example

Suppose that A, B and C are three independent events such that $P(A) = 1/4, P(B) = 1/3$ and $P(C) = 1/2$. (a) Determine the probability that none of these three events will occur; (b) Determine the probability that exact one of these three events will occur.

Solution. (a) Since A, B and C are three independent events,

$$P(\overline{A}\,\overline{B}\,\overline{C}) = P(\overline{A})P(\overline{B})P(\overline{C}) = (1-1/4)(1-1/3)(1-1/2) = 1/4.$$

(b) By additivity of probability and independency,

$$\begin{aligned} & P(A\overline{B}\,\overline{C} \cup \overline{A}B\overline{C} \cup \overline{A}\,\overline{B}C) \\ = & P(A\overline{B}\,\overline{C}) + P(\overline{A}B\overline{C}) + P(\overline{A}\,\overline{B}C) \\ = & P(A)P(\overline{B})P(\overline{C}) + P(\overline{A})P(B)P(\overline{C}) + P(\overline{A})P(\overline{B})P(C) \end{aligned}$$

□

Independence of Several Events

Solution. (b) By additivity of probability and independency,

$$
\begin{aligned}
& P(A\overline{B}\ \overline{C} \cup \overline{A}B\overline{C} \cup \overline{A}\ \overline{B}C) \\
&= P(A\overline{B}\ \overline{C}) + P(\overline{A}B\overline{C}) + P(\overline{A}\ \overline{B}C) \\
&= P(A)P(\overline{B})P(\overline{C}) + P(\overline{A})P(B)P(\overline{C}) + P(\overline{A})P(\overline{B})P(C) \\
&= (1/4)(1-1/3)(1-1/2) + (1-1/4)(1/3)(1-1/2) \\
&\quad + (1-1/4)(1-1/3)(1/2) \\
&= 1/12 + 1/8 + 1/4 = 11/24.
\end{aligned}
$$

$\square$

Independence of Several Events

Example

(**Reliability analysis**) We have two systems illustrated in Fig. 1.7. Given a particular lifetime value t_0, let A_i denote the event that the lifetime of cell i exceeds $t_0 (i = 1, 2, \cdots, 6)$. Assume that the $A_i s$ are independent events (whether any particular cell lasts more than t_0 hours has no bearing on whether or not any other cell does) and that $P(A_i) = 0.9$ for every i since the cells are identical. Determine the probabilities that the systems' lifetime exceeds t_0. Which system is more reliable?

Independence of Several Events

Analysis. For the system in Fig. 1.7(a), there are two subsystems connected in parallel, each one containing three cells. In order for the system to function, at least one of the two parallel subsystems must work. Within each subsystem, the three cells are connected in series, so a subsystem will work only if all cells in the subsystem work.

For the total-cross-tied system shown in Fig. 1.7(b), it is obtained from the series-parallel array by connecting ties across each column of junctions. Now the system fails as soon as an entire column fails, and system lifetime exceeds t_0 only if the life of every column does so.

Independence of Several Events

Solution. For the first system,

$$
\begin{aligned}
& P(\text{system lifetime exceeds } t_0) \\
&= P\big((A_1 \cap A_2 \cap A_3) \cup (A_4 \cap A_5 \cap A_6)\big) \\
&= P(A_1 \cap A_2 \cap A_3) + P(A_4 \cap A_5 \cap A_6) - \\
&\quad P\big((A_1 \cap A_2 \cap A_3) \cap (A_4 \cap A_5 \cap A_6)\big) \\
&= P(A_1)P(A_2)P(A_3) + P(A_4)P(A_5)P(A_6) - \\
&\quad P(A_1)P(A_2)P(A_3)P(A_4)P(A_5)P(A_6) \\
&= 0.9^3 + 0.9^3 - 0.9^6 \approx 0.927.
\end{aligned}
$$

Alternatively,

$$
\begin{aligned}
& P(\text{system lifetime exceeds } t_0) \\
&= 1 - P(\text{both subsystem lives are } \leqslant t_0) \\
&= 1 - [P(\text{subsystem lives are } \leqslant t_0)]^2 \\
&= 1 - [1 - P(\text{subsystem lives are } > t_0)]^2 \\
&= 1 - [1 - (0.9)^3]^2 = 0.927.
\end{aligned}
$$

Independence of Several Events

Solution. For the second configuration,

$$
\begin{aligned}
& P(\text{system lifetime exceeds } t_0) \\
&= P\big((A_1 \cup A_4) \cap (A_2 \cup A_5) \cap (A_3 \cup A_6)\big) \\
&= P(A_1 \cup A_4) \cdot P(A_1 \cup A_4) \cdot P(A_1 \cup A_4) \\
&= \big[P(A_1) + P(A_4) - P(A_1 A_4)\big]^3 \\
&= (0.9 + 0.9 - 0.9 \cdot 0.9)^3 \approx 0.97
\end{aligned}
$$

In reliability theory, we call the probability $P(\text{system lifetime exceeds } t_0)$ the reliability of the system. Since the reliability of the second system is larger than the first, the second system is more reliable. $\square$

Section 1.6 Independent of Events

- 1.6.1 Independence of Two Events
- 1.6.2 Independence of Several Events
- 1.6.3 Bernoulli Trials

Bernoulli Trials

Consider the following experiments:

- Flipping a coin and observing which side will appears.
- Rolling a die and observing whether the number six appears.
- In conducting a political opinion poll, choosing a voter at random to ascertain whether that voter will vote "yes" in an upcoming referendum.

All above examples have the same character: the experiment E that has *only two outcomes*. A ***Bernoulli experiment*** E is such kind of random experiment, the outcome of which can be classified in but one of two mutually exclusive and exhaustive ways, mainly, *success* (denoted by S) or *failure* (denoted by F) (e.g., female or male, life or death, nondefective or defective).

Bernoulli Trials

A sequence of Bernoulli trials E_n occurs when a Bernoulli experiment is *performed several independent times* so that *the probability of success*, say, *p, remains the same* from trial to trial. That is, in such a sequence we let p denote the probability of success on each trial. In addition, frequently $q = 1 - p$ denote the probability of failure; that is, we shall use q and $1 - p$ interchangeably.

Theorem

The outcomes of E_n *are the* 2^n *sequences of length* n*. The number of outcomes of* E_n *that contain* a *exactly* k *times is given by the binomial coefficient* $\binom{n}{k} = \dfrac{n!}{k!(n-k)!}$.

Bernoulli Trials

Theorem

The probability that the outcome of an experiment that consists of n Bernoulli trials has k successes and $n-k$ failures is given by:

$$P_n(k) := P_n(\#S = k) = \binom{n}{k} p^k q^{n-k}.$$

Here $\#S$ means the number of successes.

Proof. Now, suppose that, when we performed the n trials, we first obtained k consecutive successes S and then $n-k$ consecutive failures F. By independence,

$$P(\underbrace{SS\cdots S}_{k\ times}\ \underbrace{FF\cdots F}_{(n-k)\ times}) = P(S)^k P(F)^{n-k} = p^k(1-p)^{n-k}.$$

Bernoulli Trials

Proof. Hence, given that we can place the k successes among the n trials in $\binom{n}{k}$ different ways, we deduce that

$$P_n(\#S = k) = \binom{n}{k} p^k q^{n-k}.$$

$\square$

Summary for Section 1.6

- A and B are independent if and only if $P(AB) = P(A)P(B)$.
- The n events $A_1, A_2, \cdots, A_n$ are **independent** if and only if for any subset $A_{i_1}, A_{i_2}, \cdots, A_{i_k}$,

$$P(A_{i_1}^* A_{i_2}^* \cdots A_{i_k}^*) = P(A_{i_1}^*)P(A_{i_2}^*) \cdots P(A_{i_k}^*), \quad (k = 2, 3, \cdots, n)$$

  where A_i^* denotes either A_i or $\overline{A}_i$ (the same on both sides of the equation).

The end

Thank you for your patience !